

Anatomy of the Russian Agreement Mishmash*

Louis-Harry Desouvrey

First draft

December 2024

louish8@gmail.com

Abstract

In this paper, I propose an analysis of agreement in the context of the genitive of quantification in Russian. Building on a theoretical framework where features take center stage, I argue that all Russian numerals but *odin* ‘one’ are inherently vectors (ω -specified), which causes them to be incompatible with accusative and nominative morphology. As a result, numerals are not declined in direct case, which leaves them without a thematic node. Therefore, their complement can only be in the genitive, which is devoid of a thematic feature. In addition, I suggest that the grammar highlights the defectiveness of those numerals by raising a warning: undeclined lower numerals (two to four) are enhanced with the feature singular [Sg], which forces their normal underspecified plural [$\emptyset$] to recess, i.e. to be deactivated. Thus the singular is mandatory in the complement of such numerals, the recessive plural showing up only when the NP is modified by an adjective. It is shown that the agreement process follows from the nature of the representation: when the AGR features within a numeral phrase pair up, the verb must take its default morphology, whereas plural agreement with the numeral is forced when the genitive noun is attached to the verb. Further agreement complications allow us to modify the theory of coreference proposed in Desouvrey (2003). It is hypothesized that the referential node, or R-node, of an NP or a pronoun is the image of their thematic node. Therefore, if such elements do not have a thematic node, they may not have an R-node, a deficiency which hinders their ability to be the antecedent of an anaphor.

Keywords: Agreement, feature pairing, thematic features, coreference, animacy effect, paucal warning, stress-shift warning, approximative inversion, distributive *po*-phrases, vector interaction, pendulum effect, underspecification.

1. Introduction

As is well known, Russian numerals do not form a uniform class. All numerals but *odin* ‘one’ appear in a syncretic morpheme in the nominative and the accusative and require their complement to be in the genitive. This phenomenon is referred to as the genitive of quantification (GQ) in the literature. In addition, lower numerals (two to four) must have their complement in the genitive singular, but a modifying adjective appears in the genitive plural (1). On the other hand the complement of a higher numerals (five and over) must be uniformly in the genitive plural (2) with no number mismatch with modifying adjectives. When a numeral phrase is in subject position, the verb shows a variable agreement, as it can be either in the plural form or the default singular form (3).

- (1) a. dva/tri/četyre stula (adapted from Asinari 2020)
two/three/four chair.GEN.SG

* Comments are welcome!

- b. dva/tri/četyre krasnyx stula
‘two/three/four red.GEN.PL chairs’
- (2) pjat' stolov / pjat' krasivyx stolov
five table.GEN.PL / five beautiful.GEN.PL table.GEN.PL
‘five (beautiful) chairs’
- (3) a. tri studenta byli na koncerte (Asinari 2020, Baylin and Nevins 2008)
three student.MASC.GEN.SG be.PST.PL in concert
‘Three students were in the concert.’
- b. tri studenta bylo na koncerte
three student-GEN.SG be.PST.NEU in concert
‘Three students were in the concert.’

In addition, if the numeral head follows the complement, it expresses what is referred to in the literature as the approximative inversion. In subject position, the approximative phrase requires the verb to take singular morphology (4).

- (4) Studentov pjat' *sdali/sdalo èkzamen. (Franks 1994)
students.GEN.PL five passed.PL/NEU.SG exam

Finally, GQ cannot be the complement of oblique verbs (5).

- (5) a. *Ja pomogaju šest' devušek (Pesetsky 1982)
I.NOM help six girls.GEN.FEM.PL
- b. Ja pomogaju šesti devuškam
I.NOM help six.DAT girls.DAT.PL

Those data are a real mishmash, and can be best accounted for in a theory that systematically uses features, as I will show. Any analysis must explain:

- (6) a. why numeral phrases have identical forms in nominative and accusative;
b. why numerals cannot have their complement in the accusative or the nominative;
c. why the head of an NP complement of a lower numeral is in the singular while its modifying adjective must be in the plural form;
d. why GQ does not occur with oblique verbs;

- e. why the verb can be either in the singular or the plural with a subject numeral phrase.

(6) and other problems will be thoroughly accounted for as I proceed. In the next section, I briefly present the basic tenets of the theory I build on. In section 3, I show that Russian numerals cannot decline in direct case, which triggers the paucal warning. In section 4, I tackle the genitive of quantification, showing that the genitive is the only match for the undeclined numerals. In section 5, I address the agreement process with both lower and higher numerals, including the animacy effect. In section 6, I extend the analysis to constructions with distributive *po*. In section 7, I conclude the paper.

2. Theoretical background

The theory that makes it possible to account for this complex system stems from Desouvrey (2000) and earlier work, which draws on nonlinear phonology and morphology. The core of this theory is the search and the use of syntactic features from which the analysis can proceed. In addition, the syntactic structure is built sequentially by merging fully inflected heads and their complements, as opposed to theories that assume a predefined syntactic structure with a wealth of functional elements in which lexical elements and affixes are inserted. The merging process, which obeys strict rules (7), results either in the pairing of elements with identical features or their alignment if their features are compatible. In addition, in case of multiple choices, i.e. all three conditions (7a-d) are possible, the use of specified features is favored over underspecified ones, assuming the Blocking Principle (Aronoff 1976, Lumsden 1987, Desouvrey 2000).

(7) **Constraints on Merge** (cf. Desouvrey 2000, 2024, etc.)

A head *H* and its complement *C* can merge under either (a), (b) or (c), where *f* and *t* are features and $\emptyset$ the absence of a relevant feature, which is reanalyzed as a neutral feature.

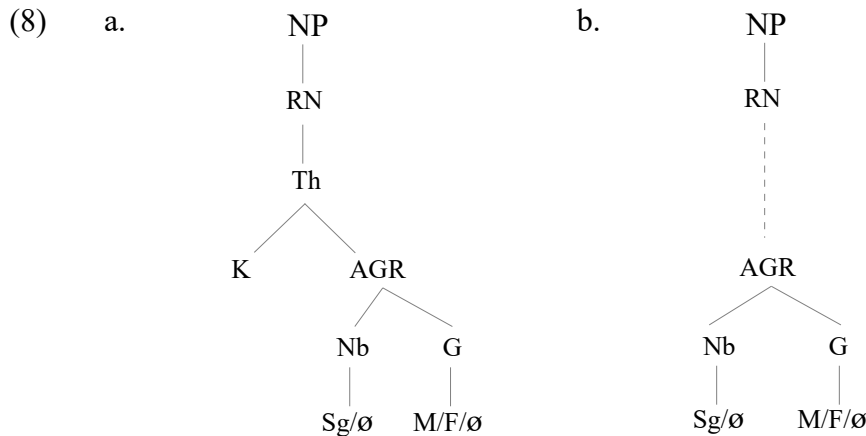
- a) Both the head and the complement have the same feature: *H* [*f*] vs. *C* [*f*]
- b) The complement is neutral: *H* [*f*] vs. *C* [$\emptyset$]
- c) Both the head and the complement are neutral: *H* [$\emptyset$] vs. *C* [$\emptyset$]

The merger of *H* and *C* cannot take place if:

- d) the head is neutral while the complement has a feature: **H* [$\emptyset$] vs. *C* [*f*];
- e) the head and the complement have different features: **H* [*f*] vs. *C* [*t*].

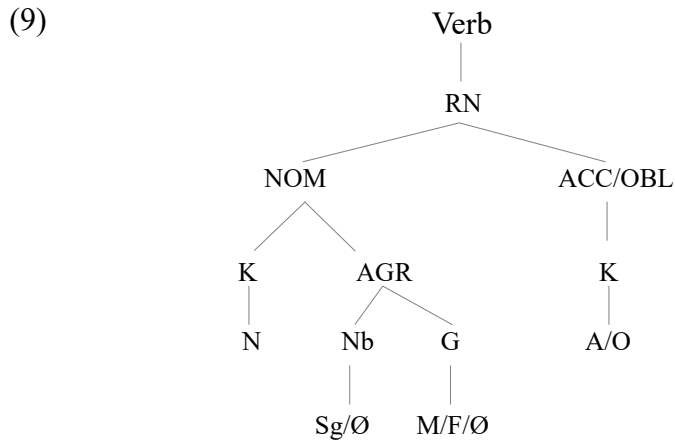
Features are hierarchically organized in a tree (8) which includes a root node (RN) which can be either $\emptyset$ (unspecified or neutral), ω (vector) or ϕ (scalar), according to the declension. Accusative NPs

have a ϕ -specified root node, while nominative NPs have a neutral root node ($\emptyset$). As for oblique declensions (at least dative, genitive, and instrumental), I assume they have an ω -root node.¹ The root node holds a thematic node (Th) standing for either NOM(inative), ACC(usative), DAT(ive), INS(trumental) or LOC(ative). It expands into two class nodes, K (case) and AGR (agreement). The latter includes two further nodes, Nb (number) and G (gender) to which the terminal features plural or singular, and masculine, feminine or neuter are attached respectively. In fact, these five terminal features are actually reduced to three, assuming the underspecification of plural and neuter, which are therefore represented with the feature [$\emptyset$]. I will assume that plural nouns lack the gender node. NPs are assigned an abstract case, either N (nominative), A (accusative) or O (oblique), through their bare K-node (8a). The genitive declension is different in that it lacks a thematic node and presumably a K-node, and thus its AGR node is floating, meaning it is not inherently attached to its root node, which is represented with a dashed line, as shown in (8b). In the syntactic derivation, the floating AGR node and the root node can be attached to the thematic node of the element they depend on.



This tree structure of nouns must be parallel to that of verbs. I assume that Russian transitive verbs roughly have the tree structure shown in (9), which includes a thematic node for the subject (NOM) and another one for the object (either ACC or OBL, i.e. DAT, GEN, INS, or LOC).² The K-node under NOM assigns abstract nominative case (N), while the K-node under the object thematic node assigns either accusative (A) or oblique (O).

1 I do not have sufficient information about locative morphology. Also, it is likely that one of the thematic nodes is redundant and therefore is underspecified ($\emptyset$).
2 Ditransitive verbs have three thematic nodes: NOM, ACC, and OBL. Accusative verbs have an AGR node holding a recessive animate feature (see below in the text).



A head and its dependent are merged to make the syntactic structure if their features are compatible. All root nodes must normally be aligned on the same tier, while lower nodes, which carry morphological information, must pair up according to (7), which results in a tree structure with multiple planes.

3. The paucal warning

In this theory, every element, including affixes, is defined by a root feature which can be either ω -specified (vector), φ -specified (scalar) or $\emptyset$ -specified (neutral). It appears that in Russian accusative morphology is φ -specified, while nominative morphology is $\emptyset$ -specified; I assume that any oblique morphology (dative, genitive, and instrumental) is ω -specified. The root features φ and ω are mutually exclusive, meaning they cannot interact with each other, and they may not be adjacent on the same tier (see Desouvrey 2024). On this view, the concatenation of a stem and a thematic, or case, affix can only take place if they have compatible root feature, basically $\emptyset$ vs. $\emptyset$, φ vs. $\emptyset$ or ω vs. $\emptyset$.

Based on this requirement, I would like to suggest that Russian numerals, except *odin* ‘one’, have an ω -specified stem. Such elements are thus incompatible with accusative and nominative morphology, which are φ -specified and $\emptyset$ -specified respectively. In effect, assuming that the stem and the thematic affix are concatenated in the morphology component, which includes (7), the features of the stem and that of the accusative affix clash ($*\omega - \varphi$) under (7e). Similarly, the nominative affix ($\emptyset$), which is assumedly the head, cannot merge with an ω -complement ($*\omega - \emptyset$) under (7d). As a result, the numeral is not declined, entering the syntax with an empty thematic slot X, meaning that the slot exists, but it does not contain an actual feature, as shown in (10a,b). They are thus different from genitive nouns which lack the slot altogether. Other thematic morphology, namely dative, genitive, instrumental, and locative, concatenate normally with numerals under the assumption that they are ω -specified (10c-e).

Numeral *odin* ‘one’ is compatible with every thematic morphology, and therefore one can assume that its stem is neutral (11).

- (10) a. *Numeral(ω) + ACC(φ) $\rightarrow$ Numeral.X (undeclined)
 b. *Numeral(ω) + NOM ($\emptyset$) $\rightarrow$ Numeral.X (undeclined)
 c. Numeral(ω) + DAT(ω) $\rightarrow$ Numeral.DAT
 d. Numeral(ω) + GEN(ω) $\rightarrow$ Numeral.GEN
 e. Numeral(ω) + INS(ω) $\rightarrow$ Numeral.INS
- (11) a. *odin*($\emptyset$) + ACC(φ)/NOM($\emptyset$) $\rightarrow$ *odin.AC/NOM*
 b. *odin*($\emptyset$)/DAT(ω)/GEN(ω) /INST(ω) $\rightarrow$ *odin.DAT/GEN/INST*

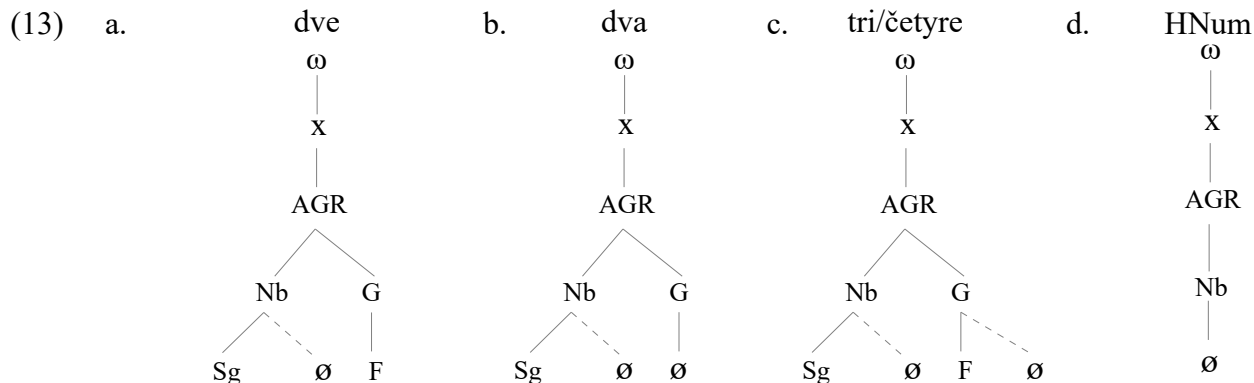
As a result of being incompatible with both accusative and nominative morphology, numerals are not declined in direct case, entering the syntax with the sole properties of their stem, namely an AGR node with a $\emptyset$ -plural specification and an empty thematic slot (X). The peculiar use of undeclined elements in an otherwise rich morphological system, I claim, prompts the grammar to raise an exception as a warning to its learners. This is realized by virtue of the following principle, whose effects have been first observed in French (see footnote 9).

(12) **The Warning Principle**

The grammar raises an exception whenever it fails to apply a constraint.

By virtue of this principle, undeclined numerals from two to four are endowed with an additional number feature, singular [Sg], besides the normal underspecified plural feature [$\emptyset$]. Thus the number node (Nb) of lower numerals comes to hold two opposite features: the singular [Sg], which becomes the dominant feature, and the underspecified plural [$\emptyset$], which is thus forced to recess, meaning it remains floating, hence inactive. Lower numerals show further complexities. In effect, the numeral *dva* ‘two’, unlike other lower numerals, distinguishes gender: *dva* is used with both masculine and neuter nouns, while *dve* is strictly used with feminine nouns. Based on this fact, I assume that *dve* is specified for feminine [F], while *dva* is underspecified [$\emptyset$] for gender, as shown in (13a,b). It is unlikely that the gender distinction is arbitrarily restricted to one numeral. I am led to take the numeral *dva* to be a beacon in the paradigm, meaning it tells us that other lower numerals do have a gender node as well. I assume that numerals *tri* ‘three’ and *četyre* ‘four’ have a gender node holding two features: feminine [F] and a recessive [$\emptyset$] for both masculine and neuter, as shown in (13c). If this is correct, the merging of a lower numeral head and a masculine noun must give rise to an insuperable (7d) effect to be

discussed below.



On the other hand, higher numerals (5+) have only one feature specification, the underspecified plural $[\emptyset]$, as shown in (13d). On this view, there is no paucal per se in this system. If paucal were intended to be a distinctive number in modern Russian, it would be available across all declensions, meaning there would be for instance a dative paucal besides dative singular/plural (but see Asinari 2020, Pesetsky 2010, etc. for another view). Rather the appearance of the singular in the NP head complement of the numeral is a warning to the learners (also see below).

In the light of this proposal, I turn in the next section to the genitive of quantification. The fact that an NP complement of a numeral has to be in the genitive will be shown to follow from the general principle stated in (7).

4. Genitive of quantification

It has been proposed in the literature that Russian numerals in direct case assign a special genitive referred to as genitive of quantification (cf. Babby 1987, Franks 1994, Rakhlin 2003, Bošković 2006, etc.), which is different from the genitive assigned by a verb or a preposition. In the present theory, only verbs and (some) prepositions assign an abstract case (either A, N, or O) to their argument, and what is traditionally taken to be case morphology is actually thematic morphology. Thus, a numeral does not assign any case to its complement; rather its lack of thematic morphology makes it compatible only with genitive.

I assume that the building of the syntactic structure obeys the following procedure. A verb picks out all the parts of its complement according to its own feature specification, and so does the head of the complement vis-à-vis its dependent, meaning the initial morphology imposed by the verb may be replaced by a more appropriate one. Thus direct case verbs (as opposed to oblique verbs) force any

numeral to take the relevant thematic node (nominative or accusative), or failing that, a compatible one. It appears that undeclined numerals are the best match for such verbs. Similarly, the numeral in turn must take a compatible complement independently of the verb. It appears that genitive, which lacks a thematic node, is the only option for the complement of X-numerals.³ As can be seen in the following table, the merger of a X-numeral with a noun bearing any other morphology yields a mismatch either in the root node or the thematic node. Specifically, a head without a thematic node cannot select a complement with a specified thematic node (cf. 7d), and an ω -specified root node is incompatible with a φ -specified root node (cf. Desouvrey 2024).⁴ I discuss in turn lower numerals and higher numerals, including their special agreement pattern.

Table 1. Direct case numerals and complement types

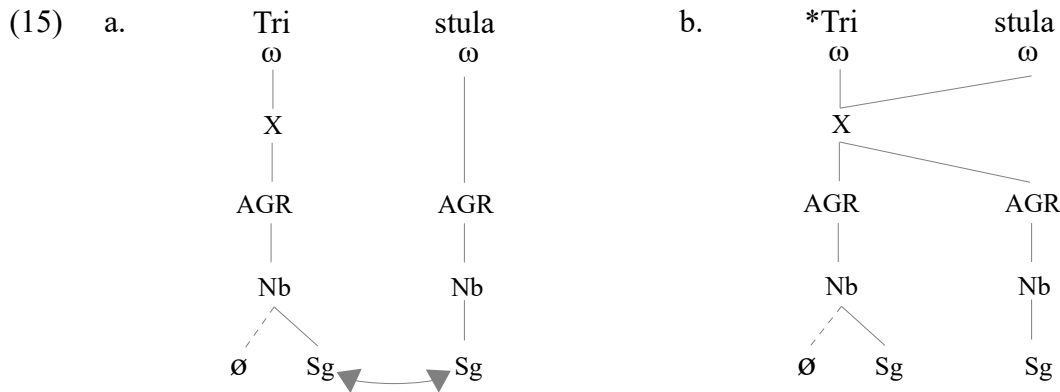
	Numerals	Head Nouns				
RN	ω	ω	$\emptyset$	* φ	ω	ω
Θ -node	X	(GEN)	*NOM	*ACC	*DAT	*INS

4.1 Lower numeral phrases

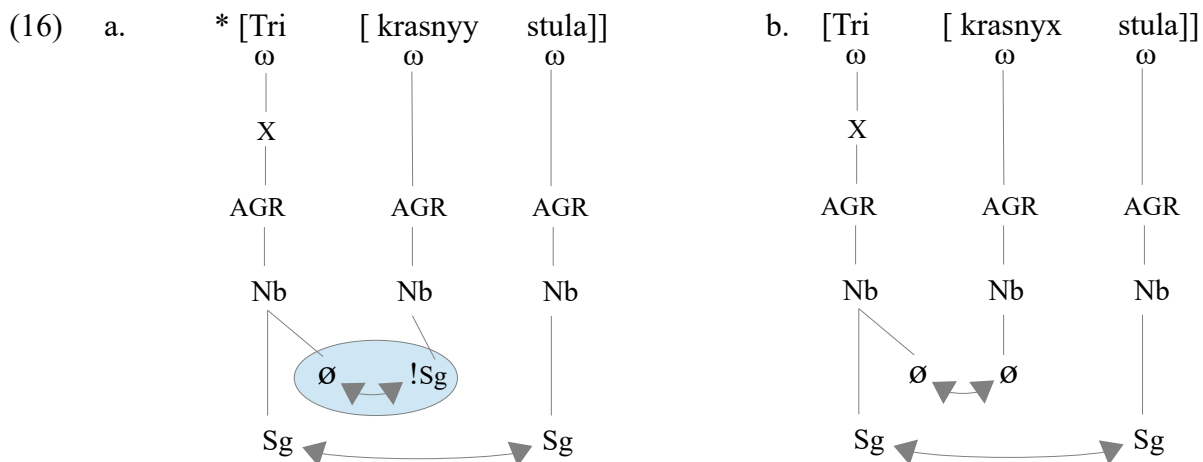
The peculiarity of lower numerals (2-4) is that they take a noun complement in the genitive singular, but an adjective modifying that noun appears in the genitive plural, as repeated in (14). Let us see how this structure is derived, given the feature trees in (13). The numeral head forces the noun to take genitive morphology, as any other case would yield a feature mismatch (cf. table 1). Additionally, since singular is the dominant feature, as opposed to the recessive feature (floating and underspecified), the noun must be in singular in order to perfectly match the numeral. Thus the singular feature of the noun pairs up with that of the numeral, as conveniently indicated with the double arrow line in (15a). It is important to note that normally the features of a genitive noun are attached to their root node in the derivation, unless they can attach to the thematic feature of another element. However, I assume that such an attachment is not possible with undeclined numerals since X is a slot, which does not contain an actual thematic node (15b). This assumption will be crucial for the account of certain types of agreement.

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- 3 I assume that the numeral is the head, as it imposes genitive morphology on the noun. If the noun were the head, it would have to mutate into genitive since accusative morphology is incompatible with the ω -node of the numeral (Desouvrey 2024). And then the genitive noun would force the numeral to be in genitive, its perfect match. See Corbett (1993) for an interesting discussion of the notion of head in the context of Russian numerals.
- 4 The incompatibility between root nodes is not crucial here. In certain context, it can generally be avoided with a buffer, a neutral feature [$\emptyset$] generated between φ and ω , as discussed in Desouvrey (2024).

- (14) Tri (krasnyx) stula.
Three (red.GEN.PL) chair.GEN.SG



Now let us see what happens when the noun is modified by an adjective. The numeral still forces the genitive on the adjective and the noun, which in turn upholds the genitive form of the adjective. Once the numeral is merged with the modified noun, its recessive plural is activated in order to pair up with the number feature of the adjective. If the adjective is in singular, an unwanted (7d) effect results in, as shown in (16a) (the exclamation sign in front of a feature indicates a mismatch). As a solution to the problems in (16a), the singular noun must instead impose underspecified plural morphology to the adjective, consistent with (7b), which implies a transitory disregard of the blocking principle. Then the NP is merged with the numeral, and as a result the Sg of the genitive noun pairs up with that of the numeral, while the adjective perfectly pairs up with the recessive plural feature of the numeral (16b). Under this analysis, there is no feature mismatch in such phrases in Russian. (The transliteration of the adjective in (16a), obtained from Google Translate, may be different from that of the other examples.)



The agreement patterns in (16), where the genitive is the only option for the adjective, concerns

masculine nouns. If the noun is feminine, the adjective can be either in the genitive plural or the nominative plural (the more common variant) (Asinari 2020, Corbett 2010, Pesetsky 2013). This is exemplified in (17).

- (17) a. dve krasivyx lampy (adapted from Asinari's (8))
 two.FEM beautiful.GEN.PL lamp.FEM.GEN.SG
 b. dve krasivye lampy
 two.FEM beautiful.NOM.PL lamp.NOM.PL
 ‘two beautiful lamps’

With respect to this state of affairs, the questions arise as to why feminine nouns behave differently from other nouns, and why the use of the nominative plural is preferred, even mandatory if more than one adjective is used. I argue that the appearance of the nominative plural besides the genitive singular is driven by economy, and it is possible because these two declensions are mostly syncretic, as mentioned in the literature. As a matter of fact, if there is a formal difference between them, the alternation is excluded, as in the case where the nominative plural and the genitive singular show a different stress pattern (cf. Pesetsky 1982, footnote 27, referring to Borrás and Christian 1971).⁵ Moreover, the nominative alternation is impossible with higher numerals, as the nominative plural is not syncretic with the genitive plural declension of the head noun, and the economy consideration is irrelevant.

Let us take a closer look at the representations in (16). They are closely bound together so that (16a) is a necessary step in the derivation of (16b). In effect, the head noun must select the singular adjective, its best match, without consideration for its eventual merger with the numeral, and then the structure fails. I assume that such a derivation iterates with a ban on the genitive singular adjective. The insuperable false start in (16a) and the iteration it triggers are costly, assuming an economy principle is active in the grammar. Such a pattern can be avoided with feminine nouns.

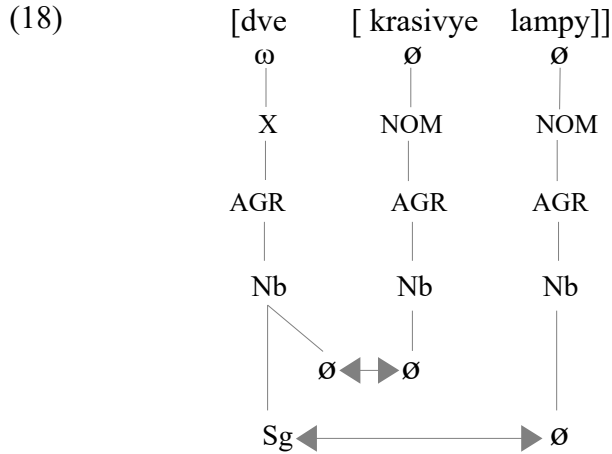
Consider the derivation of the nominative variant in (17b). In order to avoid a false start as in

5 In examples provided by Asinari (2020), the noun must be in the genitive, even if it is differentiated by stress from the nominative, while the adjective can be either in the nominative or in the genitive:

(i) dve širok-ix /širok-ie rek-í / * rék-i (Asinari's (60))
 two wide-GEN.PL/NOM.PL river.GEN.SG/*NOM.PL
 ‘two wide rivers’

Pending further information, I will assume that if the genitive singular is different from the nominative plural, the adjective cannot be in nominative, following Pesetsky's footnote 27, which is consistent with our assumptions. In effect, a genitive head, which lacks a thematic node, normally cannot be the head of a nominative adjective (cf. 7d).

(16a), the noun takes the nominative plural morphology, tricking the numeral head thanks to its syncretism with the genitive singular. Then the noun normally picks out the nominative plural adjective, which is its perfect match, eliminating the transitory disregard of the blocking principle (cf. 16b). When the numeral and the NP merge, their features correctly interact, yielding two pairs, [$\emptyset - \emptyset$] and [Sg – $\emptyset$], as shown in (18).



In addition, if there are more than one adjective, the genitive plural cannot be used, as exemplified in (19) (cf. Asinari 2020). The problem with (19b), I argue, is an effect of the pendulum property of vectors (20), which yields a structure with multiple states, or phases. In an ideal structure, all the states must be equally acceptable; otherwise at least one state must be well-formed. In the latter case, the grammar should try to look for an ideal structure whenever possible.⁶ Feminine nouns, unlike masculine nouns, can avoid structures with ill-formed states, as I will show.

- (19) a. tri starye belye kosti
 three old.NOM.PL white.NOM.PL bone.NOM.PL (=GEN.SG)
- b. *tri staryx belyx kost-i
 three old.GEN.PL white.GEN.PL bone.GEN.SG (=NOM.PL)
 ‘three old white bones’

(20) **Pendulum Principle**

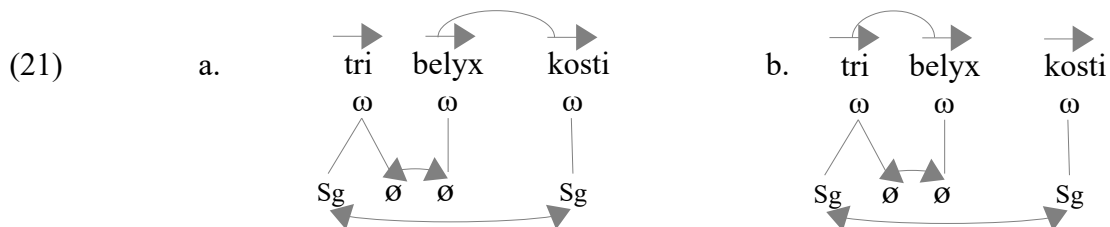
Vectors alternately combine (or pair up) with one another, yielding a multiple states structure with the following properties:⁷

⁶ This might be subject to variation across languages: certain languages may accommodate with one good state, while other may reject the whole structure if one state fails.

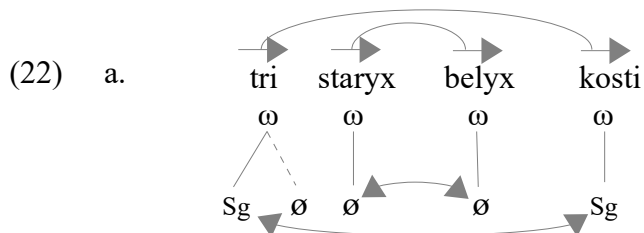
⁷ The Pendulum Principle may have other unknown properties.

- a) each state consists of different pairs of vectors and may include a free vector;
- b) a free vector can interact with vectors in a pair;
- c) vectors in a pair cannot interact with other paired vectors.

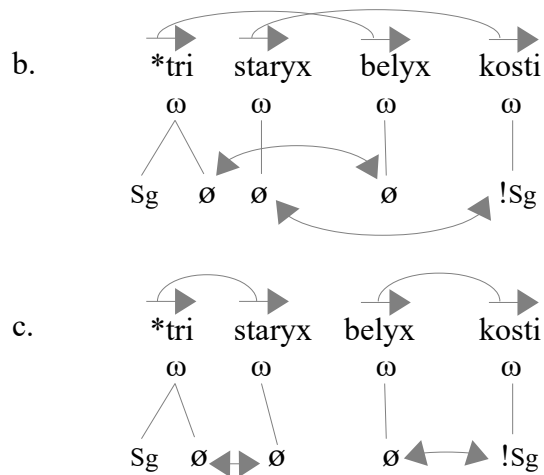
Let us first observe the effect of (20) on a simple phrase with one adjective like *tri belyx kosti* ‘three white bones’ (cf. 19b). The three ω -specified elements alternately pair up, yielding two states with one free vector, as shown in (21) (class nodes are omitted).⁸ In one state the number features of the free vector perfectly pairs up with each element in the pair of vectors (21a), while in the other the dominant feature of the numeral pairs up with the singular feature of the free vector (21b).



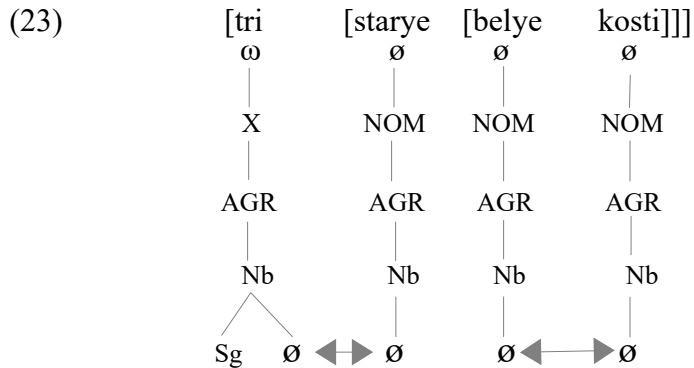
However, if another adjective is added (19b), not all states of the sentence can be felicitous. There are three possible states with no free vector, as shown in (22). In one state, (22a), the numeral and the noun make up a pair, and as a result their singular features normally pair up; the two adjectives are in another pair and their plural features pair up. However, the two other states in (22a,b) are not acceptable. A basic property of vectors is that a higher vector has scope over a lower vector, which implies that the left-hand vector in a pair has scope over its mate. So it must be the case that the left-hand vector behaves like a head, hence the pairing within a pair is subject to (7d). Therefore when any adjective combines with the noun, a (7d) effect obtains, causing the state to crash.



⁸ Two vectors cannot combine across a free vector. Thus, the pair numeral-noun is excluded (cf. Desouvrey 2022). More generally a structure with three vectors has two states instead of three (see below the Barrier Effect).

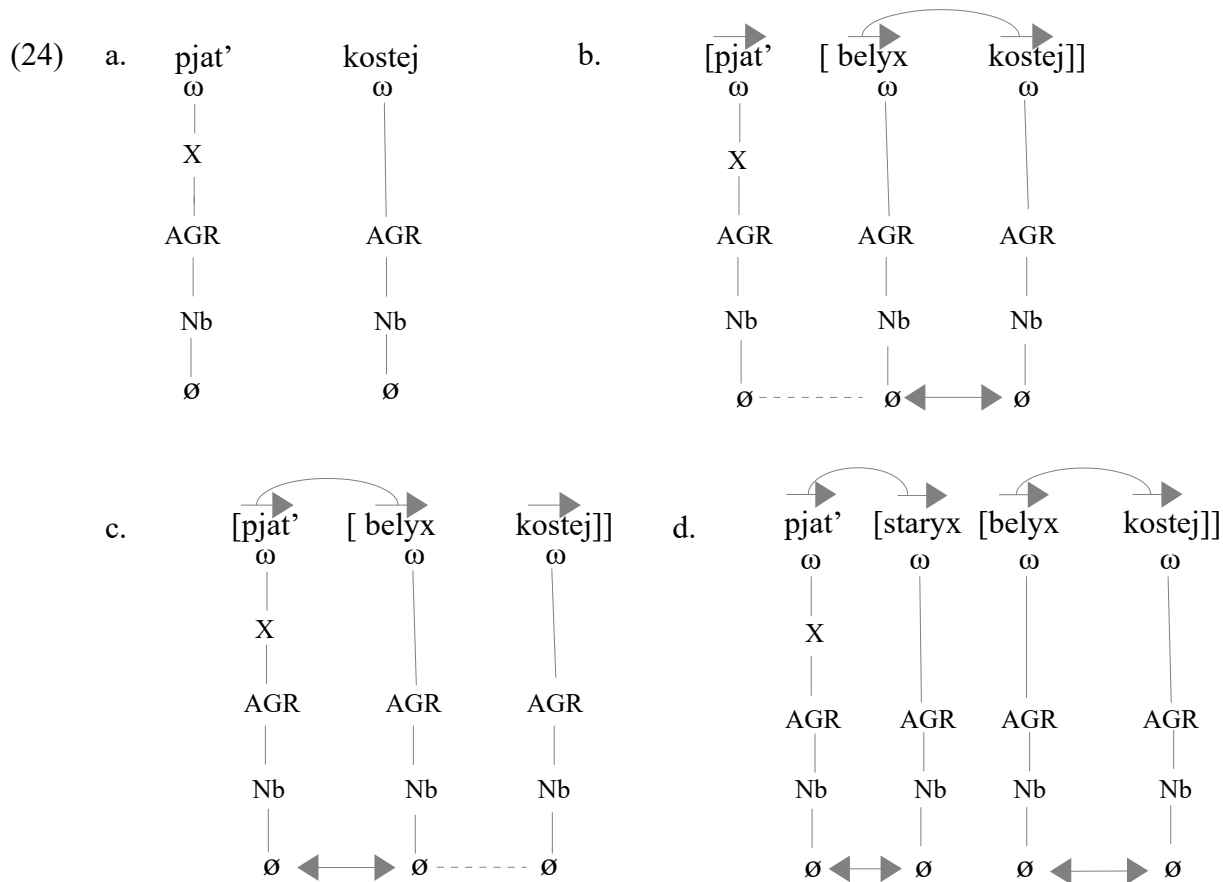


Given the syncretism of the genitive and the nominative plural, the problems in (22) can be avoided by ‘smuggling’ the nominative plural into the structure. Thus, the numeral remains the only vector in the structure, and therefore there is no pendulum effect, as shown in (23).



4.2 Higher numeral phrases

Having only one number feature (underspecified plural) (cf. 13b), higher numerals must have their NP complement uniformly in genitive plural. Thus in (24a) both number features pair up, whereas in (24b), which has two states, either the noun and the adjective or the numeral and the adjective group together, as shown in (24b,c) respectively. The stray plural feature of any free element aligns with the pair. With two adjectives, the plural features of each pair of vectors correctly pair up in each state, as shown in (24d), illustrating one state.



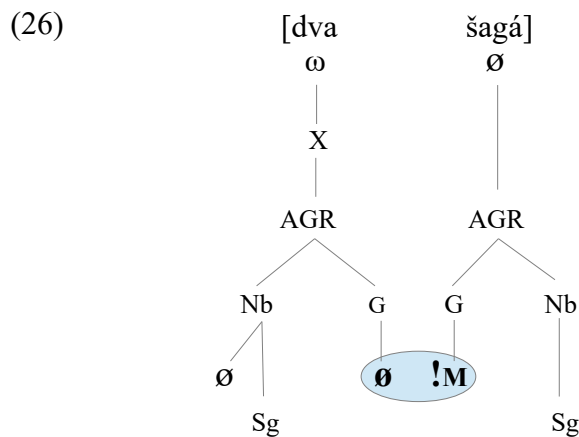
4.3 Gender discrepancy

There are five masculine nouns in Russian that undergo a stress shift in the context of the genitive of quantification: *rjad* 'row', *šar* 'sphere', *sled* 'trace', *čas* 'hour', and *šag* 'step' (cf. Asinari 2020, Pesetsky 2010, etc.). In non-quantificational environment, the stress falls on the stem (25a), while in GQ the stress moves to the declension affix (25b).

- (25) a. s pervogo šága (adapted from Asinari's (18) and (19))
 since first.GEN.SG step.GEN.SG
 'since the first step'
- b. dva šagá
 two step.GEN.SG
 'two steps'

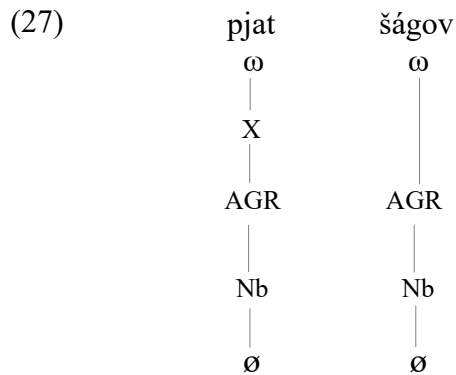
Given the feature trees discussed above and the principles governing the merging process (cf. 7), I show that this phenomenon results from the discrepancy between the features of the numeral and the

noun, which are forced to pair up despite (7d). As can be seen in the following representation (26), the numeral head has an unspecified gender feature, which opposes the masculine feature of the noun. In such a case, the pairing is normally impossible. However, since there is no other alternative for this basic and essential structure, (7d) must be overridden. I claim that the grammar raises a further warning by moving the stress from the stem to the declension affix of those five words, which serve thus as a beacon in the paradigm. One may note that the fact that all of these nouns are masculine support this analysis.⁹



On the other hand, plural nouns do not show this stress alternation. This is expected given the fact that higher numerals, which take a plural complement, do not have to distinguish gender, hence presumably lacking the gender node, just like plural nouns. As can be seen in (27), the numeral and the noun perfectly pair up with their number feature. If it turns out that plural nouns have a gender node, this would have no incidence on the analysis; such a node would stray, as it does not interact with a counterpart in the numeral.

9 The fact that the grammar acknowledges its sporadic inadequacy by using some morphological or phonological quirks, mostly a unusual stress pattern somewhere in a given paradigm, was first observed in French. Pronominal clitics in French must exit the VP domain under the Obligatory Contour Principle (OCP). However, the OCP fails with positive imperative verbs, as there is no landing site for the clitic out of the VP (cf. Desouvrey 2000). This failure of the OCP is signaled by the blocking of elision, which is normal or mandatory in non-imperative contexts. For instance, clitic *le* 'him, it' is normally reduced to *l* as in *vous l(e) prenez* 'you take it' and *vous *le/l'avez pris* 'you have taken it', but in imperative elision is blocked as the clitic, normally incompatible with stress, is stressed out: *prenez-le!*/**prenez-l!* 'take it'.



5. Verb agreement

Agreement in Russian may be optional or obligatory. I will discuss both types of agreement in turn, showing that they follow from the general principles governing the interaction of features, as stated in (28).

(28) Principles of Feature Interaction

- a) When two features pair up, they are locked in a unique plane formed with their mother node and they need not interact with other features.
- b) A stray feature F can align itself with a pair [F – F].
- c) A stray feature T normally cannot interact with a pair of features [F – F].
- d) Two different stray features originating from two identical nodes clash.
- e) A verb cannot agree with a genitive noun attached to it.
- f) Numerals in direct case do not have a thematic node, hence genitive nouns cannot attach to them.

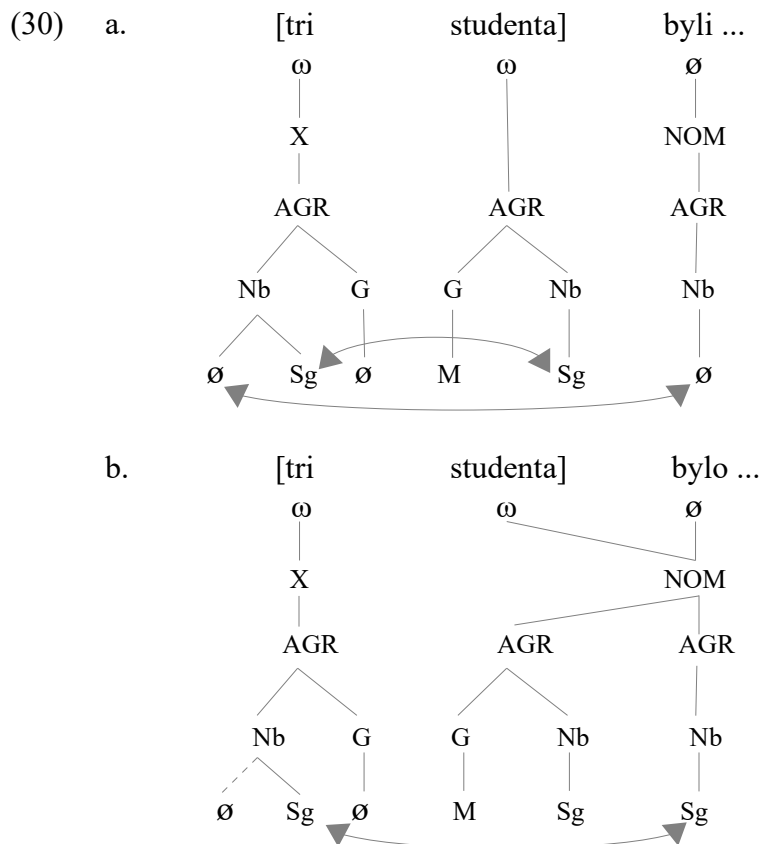
5.1 Optional agreement

When a numeral is the subject, the verb can take either plural or singular-neuter morphology, the latter being its default agreement (cf. Babby 1987, Baylin and Nevins 2008, Bošković 2006, Asinari 2020, Rakhlin 2003, Franks 1994, etc.). This is repeated in (29).

- (29) a. Tri studenta byli / bylo na koncerte.
 three student.GEN.SG be.PST.PL / be.PST.NEU in concert
 ‘Three students were in the concert.’
- b. Pjat’ devušek rabotali/rabotalo tam. (Adapted from Bošković 2006)

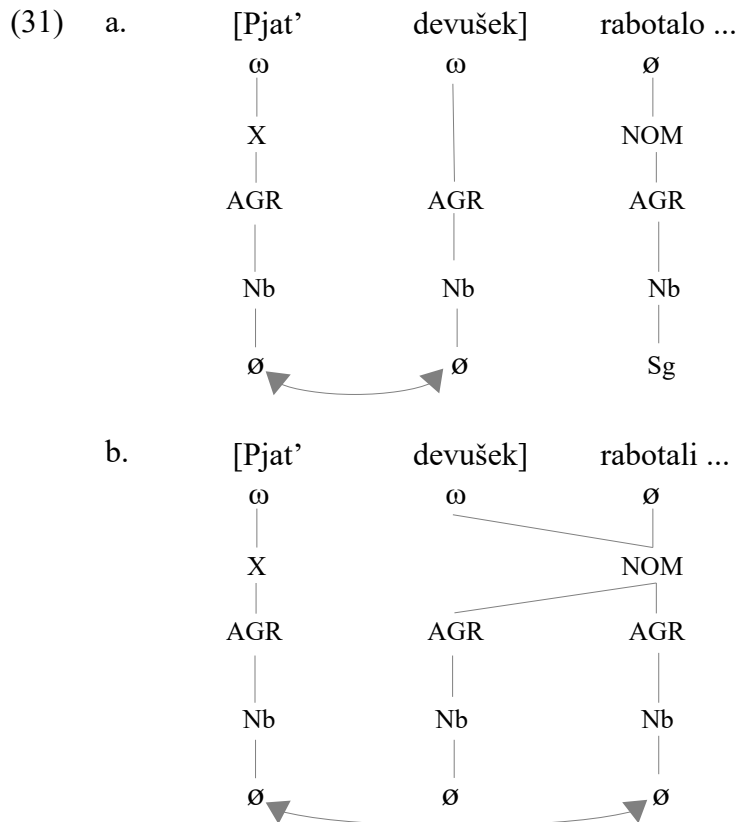
five girl.GEN.PL worked.PL/SG.NEU there
 ‘Five girls worked there.’

The interaction of the features in these structures will allow us to unravel the agreement process for both types of numerals. Consider sentence (29a). The numeral head forces the noun to be in the genitive singular, its dominant number feature, and both singular features automatically pair up. At the next step of the derivation, the verb is merged with the numeral phrase and can only agree with the remaining recessive plural feature ($\emptyset$), as shown in (30a). Alternatively, the genitive noun may attach to the verb after the merger. In such a case, the verb can agree with the dominant singular feature of the numeral. The recessive plural feature need not interact with the Sg attached to the verb and remains dormant, or else it can normally pair up with the singular attached to the verb without inducing a (7d) effect, as the singular becomes a feature of the verbal head. It turns out that the singular marking of the verb in (30b) is due to its agreement with the dominant feature of the numeral, not to its default morphology.



Consider now the case of higher numerals, which do not have a singular feature, as in (29b). The plural feature of the numeral pairs up with that of the noun, and as there is no remaining feature, the

verb normally takes its default agreement morphology, neuter singular (31a). On the other hand, if the agreement features of the genitive noun attach to the verb, agreement between the verb and the numeral takes place, as shown in (31b).



To conclude, with lower numeral phrases the verb can agree either in plural or in singular with the numeral head. With higher numerals, the verb can agree in plural with the numeral head if it holds the genitive AGR node; the singular option, which is its default morphology, occurs when the pairing in the numeral phrase is upheld after the merger.

5.2 Non-optional agreement

In certain contexts, only one type of morphology is available to the verb. The plural morphology on the verb is obviously due to agreement with the numeral head, since there is no agreement with genitive nouns. On the other hand, the singular morphology may be due either to agreement with a quantifier head or its default morphology. Both types of agreement interact in structures with coreference, as I will show in the next subsections.

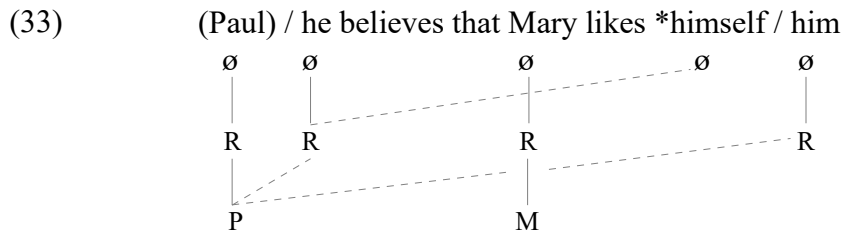
5.2.1 Plural agreement

Plural agreement appears to be the only option in the following examples (cf Franks 1994). In (32b,c), where the numeral phrase is the antecedent of an anaphor, the verb cannot be in singular; compare these sentences with (32a). Thus, apparently coreference between the subject and the object is sensitive to the agreement process. This fact, I will argue, can be explained if the thematic node is linked with the referential feature, given the theory of coreference proposed in Desouvrey (2003).

- (32) a. Pjat' ženščin smotreli/smotrelo na Ivana.
 five women.GEN looked.PL/SG at Ivan
- b. Pjat' ženščin smotreli/*smotrelo na sebja.
 five women.GEN looked.PL/SG at themselves
- c. Pjat' studentov pomogali/*pomogalo drug drugu na èksamene.
 five students.GEN.PL helped.PL/NEU.SG each other.DAT on exam

In this theory of coreference, there is a three-way distinction between the referring elements. An NP is referentially autonomous as it holds a referential feature (RF), which is the virtualization of a real world entity into the grammar. Conveniently such a feature will be represented by the first letter of that NP. This feature is attached to the NP via a referential articulation, referred to as R-node. A pronoun (third person), on the other hand, is not autonomous and must get a RF from an NP in the discourse via its bare R-node.¹⁰ Finally, an anaphor has a bare root node, and therefore must depend on an NP or a pronoun that can provide it with an R-node. Let us illustrate this with the sentence in (33). Whether the subject is *Paul* or the pronoun, it cannot spread its R-node to the anaphor, given the ban on line crossing. On the other hand, the feature *P* can spread to the R-nodes of both pronouns, whether its bearer is in the sentence or not. Notice that the RFs *P* and *M* are not syntactic features, and therefore they cannot align themselves on the same tier, which explains that *P* can spread to *R* without inducing a clash or a line-crossing effect.

¹⁰ Notice that first and second person pronouns are referentially invariable, and therefore they are represented with numerals 1 and 2 attached to the R-node.

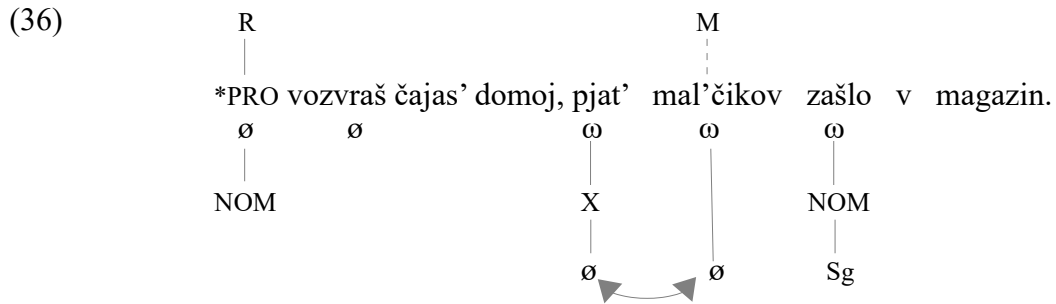


The referential representation in (33) is independent of the syntactic representation, where morphosyntactic features interact as a result of the merging of their bearers, as seen above. In the light of the data in (32), it appears that coreference is sensitive to agreement. In order to account for this fact, I will modify this theory of coreference so as to integrate it into the morphosyntactic representation. In order to preserve the essential of the representation in (33), I hypothesize that any sentence has a referential superstructure which is made of the projection of the root nodes and the thematic nodes of its referring elements. Both the root node and the thematic node of NPs and pronouns project an image, unlike anaphors which only project an image of their root node. The images of the thematic nodes are converted into R-nodes, which must align themselves on the same tier. Thus if an NP or a pronoun does not have a thematic node, it cannot have an R-node. On this view, one can say that a sentence has a morphosyntactic dimension and a referential dimension which is made of interdimensional elements.

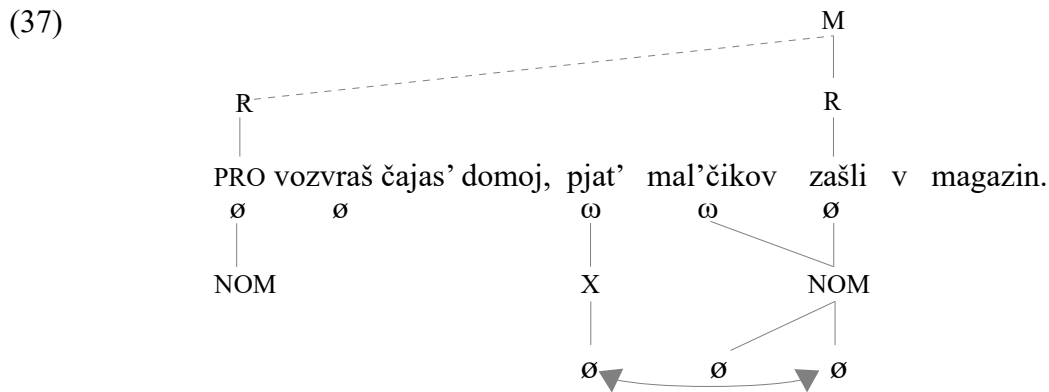
Given this hypothesis, consider the morphosyntactic representation in (34a) (cf. 32c) in which the RF *S* is added as a floating element since genitive nouns do not have a thematic node. The plural feature of the genitive noun and that of the numeral pair up, and therefore the verb must take its default singular morphology, as seen above. Another consequence is that the referential feature of the genitive noun (*S*) remains floating, as it is not supported by a thematic node. Therefore, coreference cannot be set up between the anaphor and its antecedent. Alternatively, the genitive noun can attach to the nominative node of the verb, which therefore agrees in plural with the numeral. An image of the NOM node can thus be created, allowing the RF of the genitive to be anchored. The R-node can spread to the root node of the anaphor, setting up the coreference relation (34b). Notice that for the sake of clarity, the R-node is placed above the sentence tier, but it must be taken as a branch of the root node. Also, the image of the root nodes is not shown in this representation as it is not modified in the referential structure.

- Let us turn now to the case where the controlled element is PRO, as in (35a,b) below (cf. Franks 1997: 10). The contrast between sentences (35b) and (35c), where both agreement options are possible, strongly shows the apparent dependency of coreference to the morphology of the verb.

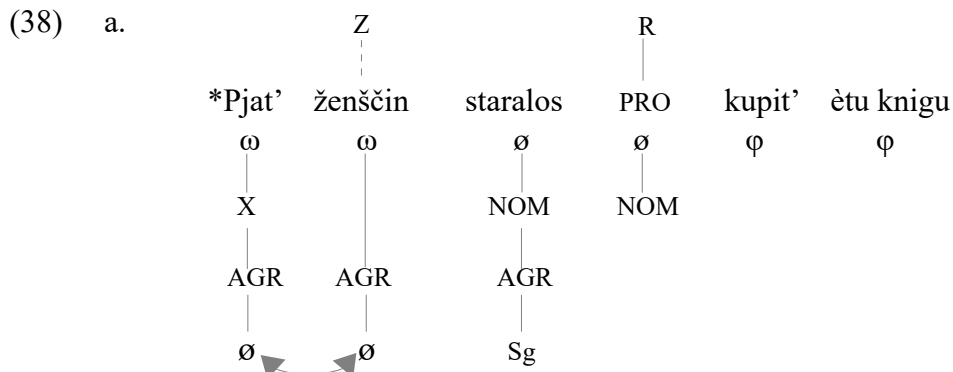
- The unacceptability of the default singular agreement in (35b) is due to the pairing of the genitive noun with the numeral head, as shown in (36). As a result, the RF of the noun cannot have an image of a thematic node which it can attach to. Thus the control of PRO, assumedly a pronoun, may not take place even though it has a thematic node which projects an image R.



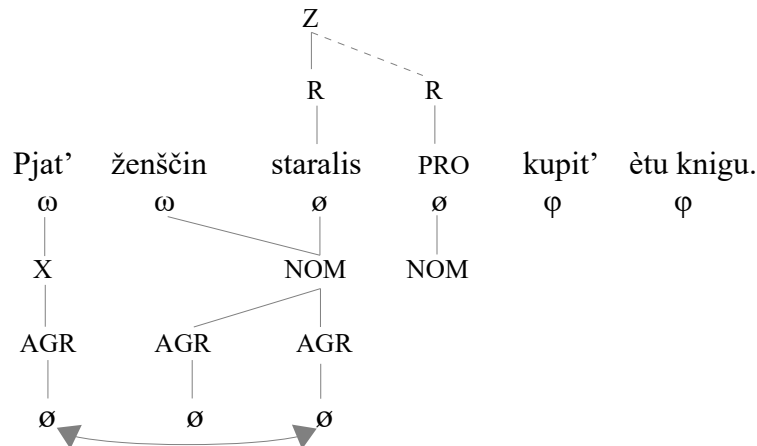
On the other hand, if the verb shares its NOM node with the genitive noun, which triggers plural agreement with the numeral, an R-node can be generated in order to support the RF of the noun. The RF *M* can thus spread to PRO, as shown in (37).



As for sentence (35a), it can be explained in the same way, which further supports the hypothesis that the R-node is the image of the thematic node. If the pairing of the genitive noun with the numeral is upheld after the merger of the numeral phrase with the verb, the RF may not have an R-node and therefore cannot control PRO, as shown in (38a). Quite contrary, if the genitive NP is attached to the verb, an R-node can be created, allowing the RF to attach itself and to spread to PRO, as shown in (38b).



b.

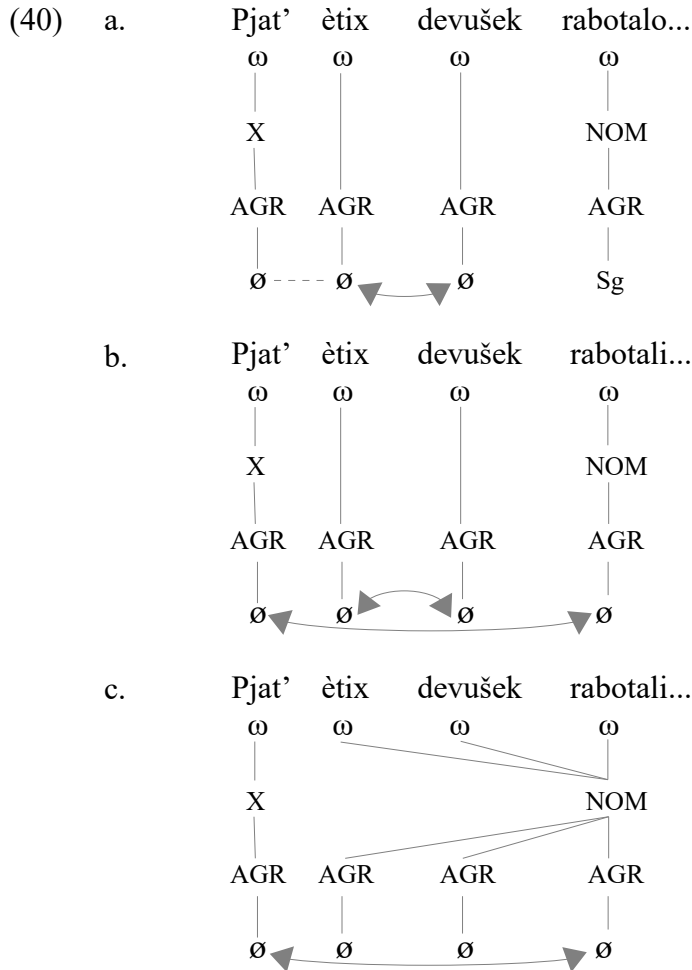


What matters in this type of structures is that the noun that antecedes a referring element must have a thematic node so that its R-node can be created. It just happens that the verb can only agree in plural with the numeral when it provides the genitive NP with a thematic node. It should be noted that since coreference is realized by spreading, it is normal for speakers to try to find an antecedent to a given referring element when judging the non-agreement examples, hence the vacillation in their judgment about these sentences, as Franks points out in his footnote 71. The essential point is that these speakers would never produce such sentences in normal speeches (i.e. not induced by linguists).

Furthermore, when a demonstrative precedes a numeral phrase, it must be in nominative, and morphosyntactic agreement, as opposed to default agreement, is obligatory; compare the examples in (39) (cf. Bošković 2006).

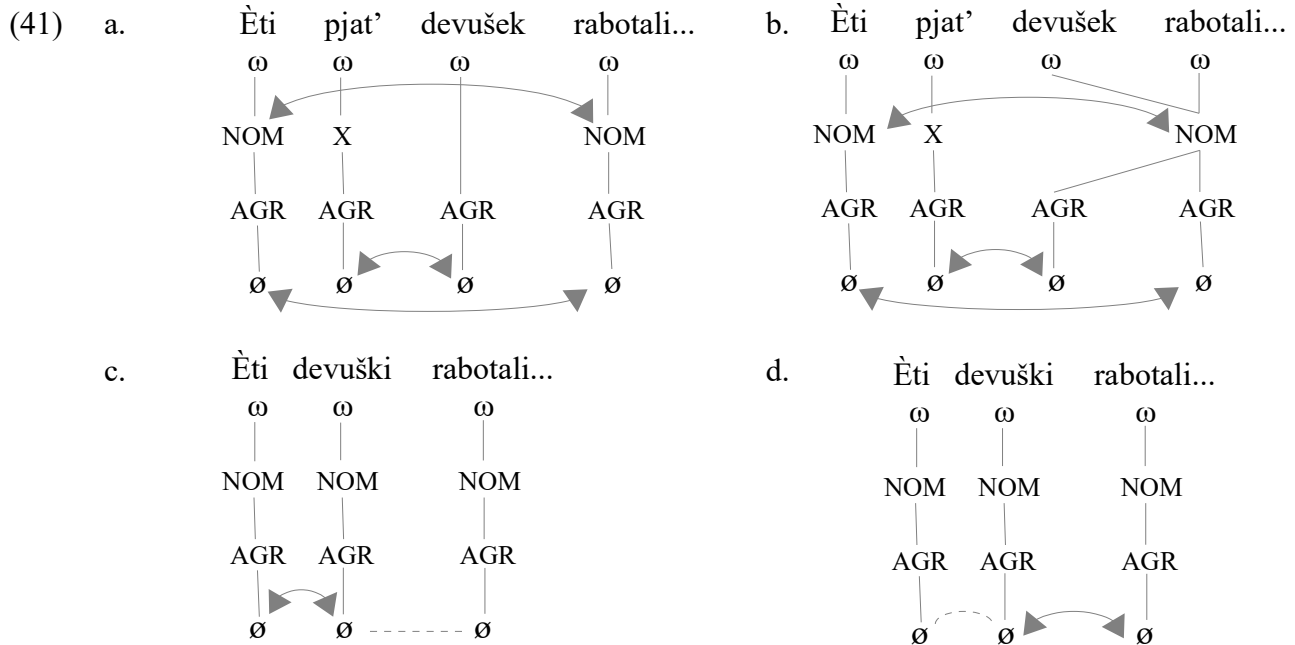
- (39) a. Èti pjat' devušek rabotali/*rabotalo tam.
 These.NOM five girls.GEN worked.PL/SG there
- b. Pjat' ètix devušek rabotali/rabotalo tam.
 five these.GEN girls.GEN worked.PL/SG there

The derivation of (39b) proceeds as follows. The numeral imposes the genitive on every element that forms its complement. The demonstrative and the noun can normally pair up, and the number feature of the numeral may or may not align itself with the plural pair, leaving the verb with two agreement options: default singular in the case of alignment (40a) or plural agreement otherwise (40b). If the genitive phrase is attached to the thematic node of the verb, plural agreement must take place (40c).



Consider now (39a). Initially, the verb tries to impose nominative on every constituent of the subject phrase. However, the numeral head can only be in its undeclined form. Then under the merging process, the numeral head forces the noun to take genitive morphology, and their AGR features pair up, making up a plane. When the nominative demonstrative enters the structure, it can align itself with the numeral phrase, as their features are compatible. Finally, when the whole phrase is merged with the verb, the nominative node of the verb and that of the demonstrative can pair up, making up their own plane, and therefore the verb permanently agrees with the demonstrative, as shown in (41a). Thus (41a) is different from (40b) in that the nominative-specified demonstrative makes a strong pair with the verb, unlike the thematic-free numeral. If the genitive noun is attached to the verb, its AGR features can still pair up with that of the numeral, while the verb upholds its agreement with the demonstrative, as shown in (41b). In a simple sentence without a numeral, every element in the phrase is in nominative as required by the verb. Thus, they can make up a pair which the verb can align with (41b). Alternatively, the verb might pair up with the adjacent noun, while the demonstrative aligns itself with the pair, as

shown in (41c).



5.2.2 Singular agreement

With quantifiers like *mnogo* ‘many’, *malo* ‘a few’, etc. singular agreement is required, even in structures with coreference, as in the examples in (42), discussed by Glushan (2013). According to Glushan, the verb must agree with such quantifiers because they are always nominative and belong to declension class IV, singular neuter.

- (42) a. Vozvraščajas' domoj, mnogo mal'čikov *zašli/zašlo v magazin
 returning home, many boys.GEN.PL dropped-in.*PL / NEUT.SING to a store
 ‘Returning home, many boys stopped by the store.’
- b. Malo studentov prišlo/*prišli na lekciju
 a few students.GEN.PL came.NEUT.SING /*PL to the lecture
 ‘A few students came to the lecture.’

These quantifiers can also be used as object. As can be seen in (43), the form *mnogo* is used as object of accusative verbs while with oblique verbs, the quantifier declines accordingly, just like numerals. I am led, therefore, to assume that such quantifiers are like numerals, i.e. they are not declinable in accusative and nominative. As for gender and number features, I assume that they are specified respectively for neuter ($\emptyset$) and singular (Sg), plus a recessive plural ($\emptyset$), which is mostly

consistent with Glushan's observation that they belong to class IV, neuter singular.

- (43) a. Ivan vstretil mnogo lingvisov (adapted from Rakhlin 2003)
 Ivan met(ACC) many linguist.PL.MASC.GEN
 'Ivan met many linguists.'
- b. Ivan vosxischen(INSTR) mnogimi lingvist.ami.
 Ivan admires many.INSTR linguist.PL.MASC.INSTR
 'Ivan admires many linguists.'

Before tackling the coreference case, one must explain why the complement of the quantifier (singular neuter) is in genitive plural in direct case environment. First of all, having no thematic node (X) the only case the quantifier is compatible with is genitive, which does not have its own thematic node. In addition, the neuter feature of the quantifier cannot pair up with most singular nouns, which have a specified gender feature, masculine (M) [$\emptyset$ – !M] or feminine (F), for which there is no warning effect. So to avoid a further (7d) effect [$*\emptyset$ – F], the complement must be in the plural, which does not distinguish gender. The quantifier being gender-compatible with neuter noun ($\emptyset$ -specified), one would expect a neuter complement to be in genitive singular [$\emptyset$ – $\emptyset$], which seems not to be the case. I assume that this is ruled out by the Harmonization Principle (44). This principle makes the grammar more learnable, as it eliminates parochial rules with various conditional implications. In the case at hand, the following parochial rule would govern the formation of such quantifier phrases: if the noun is feminine or masculine, plural must be used in order to avoid a (7d) effect; if the noun is neuter genitive singular is obligatory. Since the vast majority of Russian nouns are either masculine or feminine, the plural requirement is extended to the smaller class of neuter nouns as well. Thus by virtue of this process, the complement of this class of quantifiers uniformly takes genitive plural morphology.

(44) **Harmonization Principle**

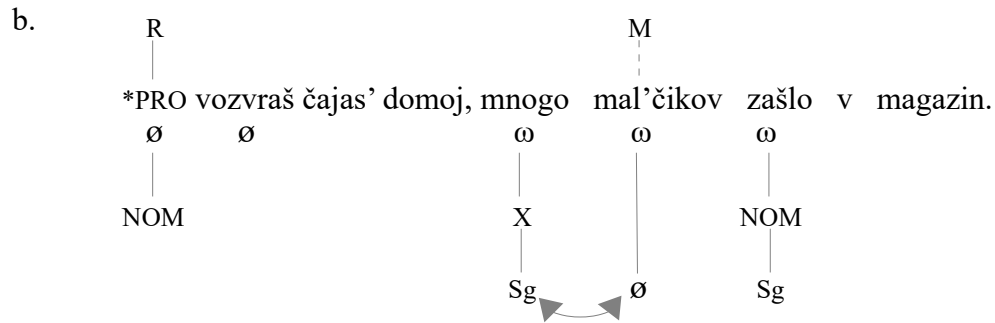
A constraint tends to overgeneralize in a given paradigm as long as no violation of other constraints results in.

Consider now (42b). When the QP is merged with the verb, the genitive noun may or may not attach to the thematic node of the verb. If the QP stays still, the verb agrees in singular with the dominant feature of the quantifier (45a). In the other alternative, the verb still agrees with the dominant feature of the quantifier, without interfering with the plural pair, as shown in (45b).

- b.
-
- Malo studentov prišlo na lekciju
- ω ω $\emptyset$ $\emptyset$
- X NOM
- AGR AGR AGR
- G Nb Nb Nb
- $\emptyset$ Sg $\emptyset$ $\emptyset$ Sg

[illegible]

- (46) a.
-
- The diagram shows a syntax tree for the sentence "PRO vozvraš čajas' domoj, mnogo mal'čikov zašlo v magazin." The root node is **R**, which branches into **PRO** and **vozvraš čajas' domoj**. Below **PRO** is the feature **Ø**, which is linked to the node **NOM**. The node **R** is also linked to a higher node **M** via a dashed line. The phrase **mного mal'čikov** is linked to a node **X**, which is linked to the feature **Sg**. The phrase **zašlo v magazin.** is linked to a node **NOM**, which is linked to the feature **Ø**. This **Ø** is linked to the feature **Sg** via a solid line. A curved arrow indicates a relationship between the two **Sg** features.



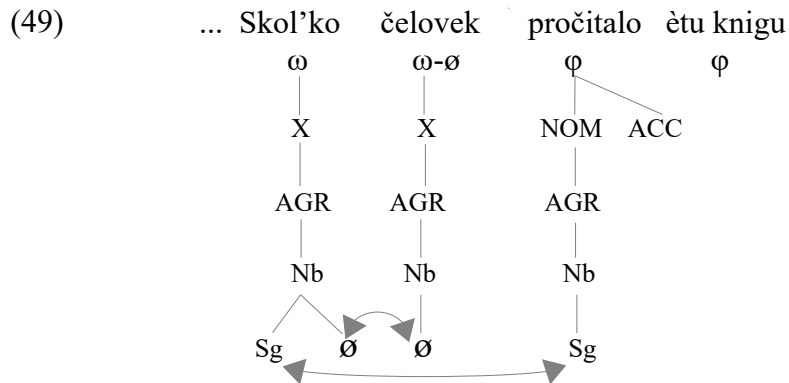
This analysis carries over to quantifier like *skol'ko* ‘how many’. Consider the following structure (47) (cf. Franks 1994). Franks assumes that *čelovek* ‘people’ is in the genitive plural, which seems to be incorrect. In effect, this word does not decline in plural and its genitive is *čeloveka* (cf. Titov 2017). This leads me to assume that *čelovek* is not declined in direct case, at least in this use, and therefore it has no thematic node, just like numerals. In addition, I assume it has a $\emptyset$ -number specification, which is not the opposite of singular.¹¹ In my view, this structure is derived from the input shown in (48), where the complementizer, which is a relative anaphor, is a constituent of the matrix clause (see for instance Desouvrey 2021, 2022, etc.). In such a structure, the quantifier always pairs up with its X-complement via its recessive plural, while the verb agrees with the remaining dominant feature, as shown in (49). From this structure, the quantifier phrase is eventually fronted. (The $\emptyset$ appended to the root node of *čelovek* is a buffer (cf. Desouvrey 2024).

- (47) Skol'ko čelovek_i [Ivan dumaet [čto [e_i pročitalo/*pročitali ètu knigu]]]?
 how-many people.∅.PL Ivan thinks that read.Nt.SG read.PL this book.ACC

- (48) [Ivan dumaet čto] [skol'ko čelovek pročitalo/*pročitali ètu knigu]?

¹¹ This word belongs to a special class of uncountable nouns, and its surrogate plural is *ljudey*. Let us assume that uncountable nouns in Russian are specified for a special feature *uC* such that *ljudey* is [*uC*] and *čelovek* is [$\emptyset$] (underspecified for *uC*). If the word *gruppa* ‘group’ is specified [*uC*], the following data (adapted from Titov 2017) makes sense, given our assumptions:

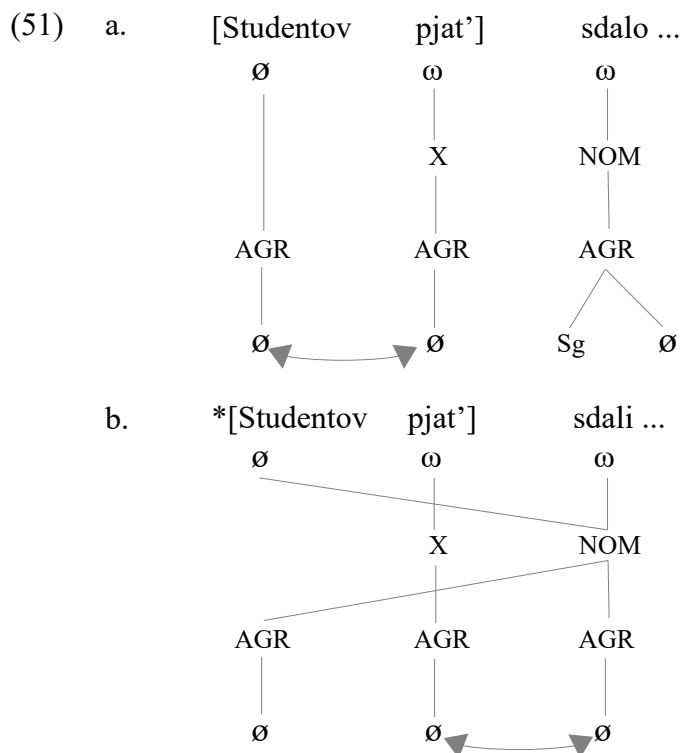
- (i) a. *gruppa(uC) ljudey.GEN.PL(uC)* vs. **gruppa(uC) čelovek.NOM(∅)/čeloveka.GEN(∅)* (blocking effect)
 “a\the group of people”
 b. **vosem'[∅] ljudey.GEN.PL[uC]* (7d effect) vs. *vosem'[∅] čelovek.NOM [∅]*
 “eight people”



A further piece of evidence for this analysis is provided by a construction where the numeral follows its genitive complement, yielding an approximative reading (50). This is known in the literature as the Approximative Inversion. In such a construction, the verb must be in singular.

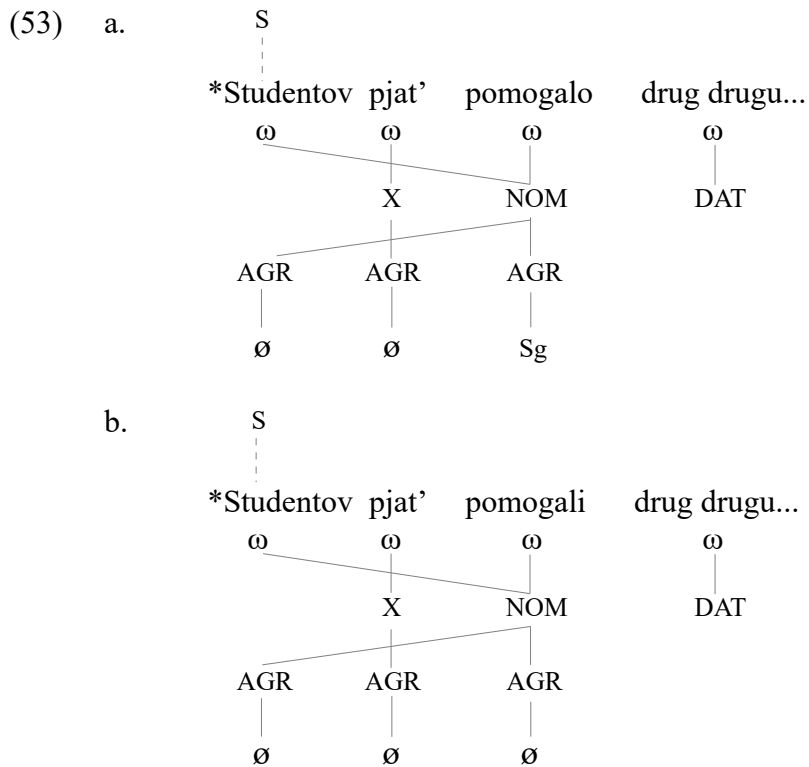
- (50) Studentov pjat' *sdali/sdalo èkzamen. (Franks 1994)
 students five passed(pl)/(sg) exam
 'About five students passed the exam.'

Here, the singular form of the verb is truly a default agreement. The genitive noun must pair up with the numeral, reducing the verb to take its default agreement morphology (51a). The plural option is not possible, as the genitive noun cannot attach to the verb without inducing a line-crossing effect (51b).



Interestingly, this analysis of (51) correctly predicts that an inverted NP cannot be the antecedent of an anaphor, irrespective of the number form of the verb, as exemplified in (52). In effect, whatever the agreeing form of the verb, the genitive cannot attach to its thematic node by virtue of the ban on line crossing, as shown in (53).

- (52) Studentov pjat' *pomogali/*pomogalo drug drugu na èkzamene. (Franks 1994)
 Students.GEN.PL five helped.PL/N SG each other.DAT on exam
 About five students were helping each other on the exam.



5.3 Animacy effect

A further complexity is brought about by animacy. Russian morphology distinguishes animate and inanimate nouns in the accusative plural. For masculine nouns of the first declension the animacy distinction extends to the singular as well (cf. Asinari 2020, Pesetsky 2013, Babby 1987, Franks 1994, Rakhlin 2003). In addition, it is pointed out in the literature that animate accusative and genitive are syncretic in both numbers, while inanimate nouns are syncretic in the nominative and the accusative. The consequence of the animacy effect is that both the numeral head and its animate complement must

take genitive morphology, as can be seen in (54a) vs. (54b) (cf. Rakhlin 2003).¹²

- (54) a. Sasha videl tri doma.
Sasha.NOM saw three house.GEN.SG
'Sasha saw three houses.'
- b. Sasha videl trjox mal'čikov /(*)tri malčika.
Sasha saw three.GEN.PL boy.GEN.PL /*three boy.GEN.SG
'Sasha saw three boys.'

In fact, this animacy effect is found in non-quantified NPs as well. This is illustrated below with three types of direct object: animate masculine, animate feminine, and inanimate masculine (cf. Babby 1987). Only the animate masculine exhibits the animacy effect (55a).

- (55) a. Ja uvidel mal'čika /*mal'čik'
I saw boy.GEN.SG.MASC /boy.ACC.SG.MASC
- b. Ja uvidel devušku /*devuški
I saw girl.ACC.SG.FEM /girl.GEN.SG.FEM
- c. Ja uvidel stol /*stola
I saw table.ACC.SG.MASC /table.GEN.SG.MASC

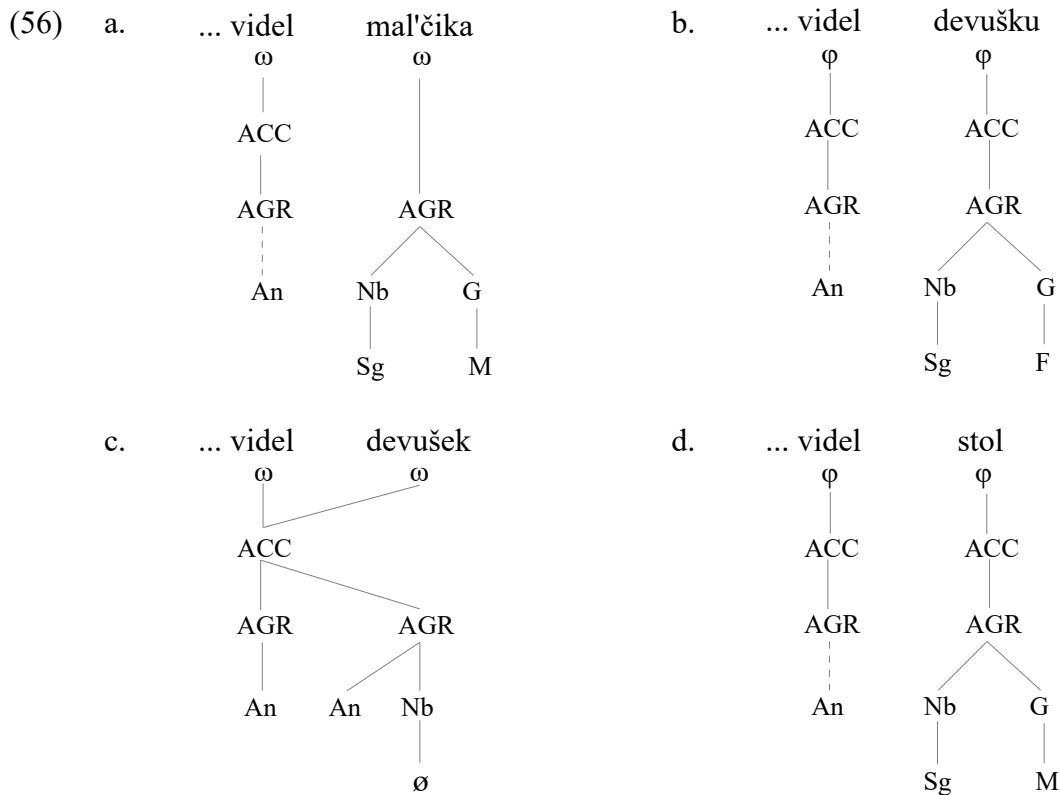
It should be noted that *mal'čik* is considered an accusative form because in the declension class 1 the accusative is expected to be syncretic with the nominative. However, since animate nouns cannot be used in accusative, one can assume that such a form like *mal'čik* (accusative masculine) does not exist. Thus, I assume that animate nouns do not decline in accusative. The genitive is used to supply the missing accusative, as it lacks its own thematic node.

In the above sentences, it is clear that the notion of animacy is crucial. So it must be the case that it is the manifestation of a feature which can be referred to as [An]. As a matter of fact, I contend that (i) such a feature is found only in the genitive plural, and (ii) it is recessive in accusative verbs, meaning it is only activated when the complement is specified for animacy. Once a new feature is found, it is a matter to locate its position in the feature tree. I suggest that [An] is what can be referred to as a wild feature, meaning it does not have its own class, and therefore it has no opposite feature,

¹² In Rakhlin (2003) and Babby (1987), the genitive of quantification is not acceptable with animate nouns, as **tri malčika* in (54b). According to Franks (1994) the animacy rule, which requires both the numeral and the noun to be in genitive, is optional with lower numerals. Hence, the asterisk in (54b) is put in parentheses. Franks's observation is consistent with our analysis (see below in the text).

specified or not (* $\emptyset$ or *In(animate)). Thus, I take it to be directly attached to the AGR node.

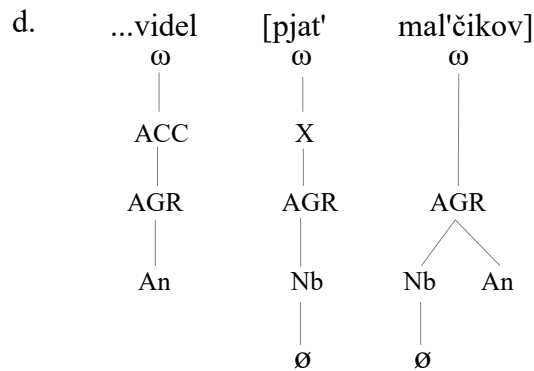
Given these assumptions, consider the sentences in (55). In (55a), the noun is in genitive because there is no accusative declension for masculine animate nouns (56a). On the other hand, an animate feminine declines normally in accusative, as shown in (56b). In (56a,b), the animacy feature of the verb is dormant since genitive singular nouns and feminine accusative nouns are not specified for animacy. If [An] were active, it would try to pair up with either Nb or G, which would give rise to a feature clash. Notice that if the animacy feature is activated in (56a), forcing genitive plural on the noun, the sentence would be perfect but with a different semantic value: the meaning would be *I saw the boys* instead of *I saw the boy*. With a plural animate feminine noun, genitive plural is the only match, as there is no accusative plural for animate nouns in all genders (56c). In this context, the animacy feature of the verb becomes active and the feminine noun is attached to the thematic node of the verb. Finally, if a masculine NP is not animate, the accusative is the best match (56d), and the animacy feature of the verb remains dormant.



Now consider the case of animate numeral phrases. Both the normal genitive of quantification and the genitive of animacy are possible (cf. 54b and footnote 12). Normally the verb picks out the elements that will constitute its complement according to its own feature specification, and then allows

c.

*videl ω	[tri ω	mal'čikov] ω
ACC	X	
AGR	AGR	AGR
	/ \	/ \
An	G Nb	Nb An
	/ \	
	∅ Sg ∅	∅



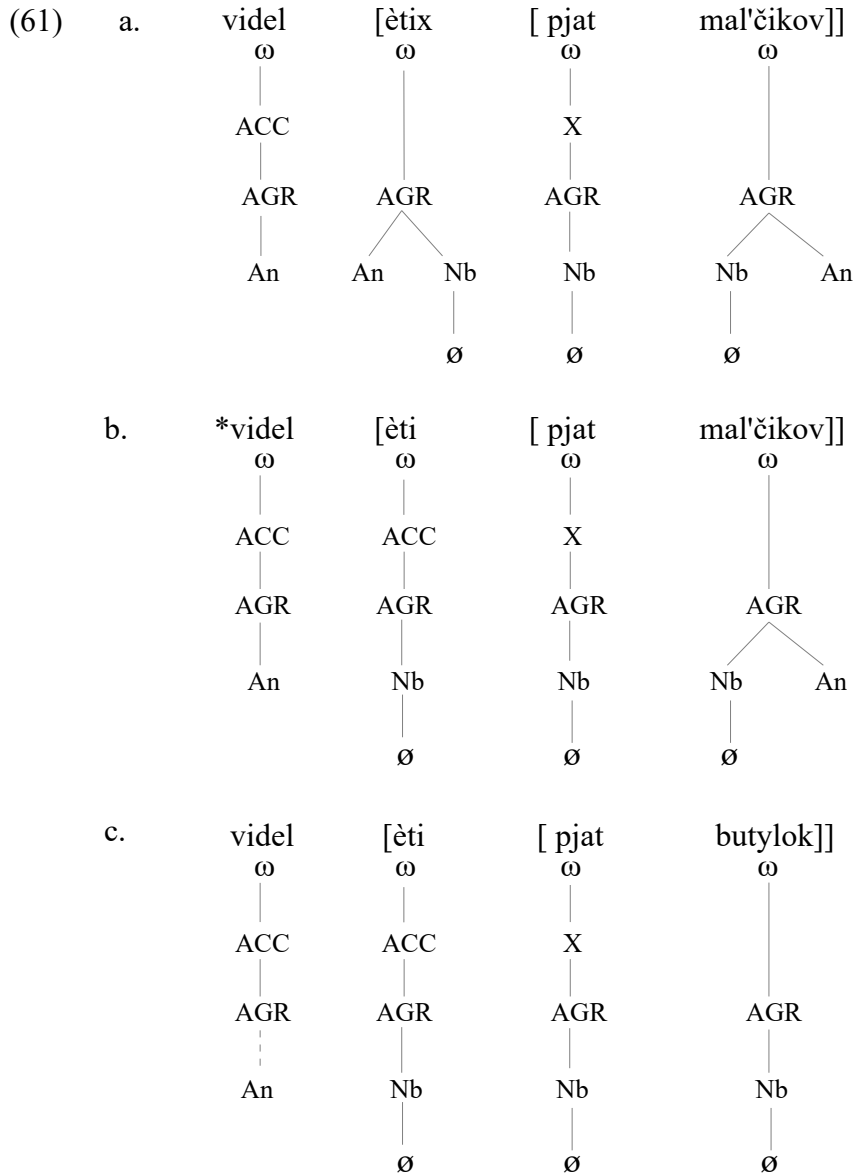
Consider now the contrast in (58), which shows that plural agreement with inanimate subjects is not fully acceptable (Glushan 2013). As discussed above, plural agreement occurs when the genitive noun is attached to the verb, which thus can only agree with the numeral head. Therefore, it must be the case that the problem is due to the attachment of the inanimate genitive to the verb. One can suppose that animacy is a property of both nominative and accusative nodes in Russian, but it gives rise to a recessive feature only in the accusative node. On this view, an inanimate noun mildly clashes with the subject thematic node it is attached to. Alternatively, it might be that animacy is slowly expanding to nominative in modern Russian by a harmonization effect. In any event, these examples support the assumption that accusative verbs have a recessive animacy feature. If the verb were not specified for animacy, this category would be just like gender in English, depending of the sole speaker knowledge of the world, without any effect on the syntax.

- (58) a. Pjat' studentov byli/bylo v Londone
 five student.SING.GEN were.PL/was.NEUT in London
 'Five students were in London.'
- b. Pjat' stuljev ??byli/bylo v Londone/v komnate
 five chairs.GEN ??were.PL/was.NEUT in London/in the room
 'Five chairs were in the room.'

Finally, consider the following data. According to Babby (1987), the noun is the head of the numeral phrase because the demonstrative modifier in (59) is in genitive, agreeing with the noun in number and animacy. In (60), on the other hand, the noun is not animate and the modifier appears in the accusative plural. If the genitive noun were the head, the numeral would be in the genitive in all numeral phrases, irrespective of animacy, and hence a sentence like (59c) would be well-formed; recall the discussion of (40) and (41) above.

- (59) a. Ja uvidel etix pjat' mal'cikov.
I saw these.GEN.PL five.X boys.GEN.PL
- b. *Ja uvidel eti pjat' mal'cikov
I saw these.ACC.PL five.X boys.GEN.PL
- c. *Ja uvidel etix pjati mal'cikov.
I saw these.GEN.PL five.GEN boys.GEN.PL
- (60) a. Ja uvidel eti pjat' butylok
I saw these.ACC.PL five.ACC bottles.GEN.PL
- b. *Ja uvidel etix pjat' butylok
I saw these.GEN.PL five bottles.GEN.PL

Let us derive these structures. The building of (59a) proceeds as follows. The verb imposes the surrogate genitive of animacy to the noun and the demonstrative, while it picks the X-form of the numeral, its best match. Then the numeral head correctly merges with the noun, upholding the initial genitive of animacy imposed by the verb. At the next step, the numeral phrase is merged with the demonstrative which is compatible with the numeral head and the genitive noun, as their animacy features can pair up. Finally, the verb merges with the whole phrase. It turns out that genitive modifier is the best match according to the blocking principle, as it can agree with the phrase in both animacy and number while still compatible with the verb, unlike an accusative modifier, as shown in (61a,b) (the demonstrative genitive eventually attaches to the verb). However, if the noun is inanimate, the animacy feature of the verb is not activated, and the verb picks the noun and the demonstrative in their normal accusative form and the X-form of the numeral. The numeral merges with the noun, forcing it to take its genitive plural form, which is not specified for animacy. The numeral phrase is then merged with the demonstrative, upholding its accusative form, as there is no mismatch in the structure. Once the whole phrase is merged with the verb, the accusative nodes of the demonstrative and the verb make up a pair which the X-node of the numeral can align with, as shown in (61c).



To conclude this section, our analysis is based on the assumption that animate nouns are not declined in accusative. The genitive is used to fill in this gap, given its lack of thematic node. The animate feature (An), which exists in the genitive plural, is dependent of the AGR node and hence is not the opposite of an inanimate feature (underspecified or not). There is no evidence that inanimate nouns need to be underspecified for animacy (∅). Notice that the fact that animate nouns are not declined in accusative might not be a simple gap in this complex system. If there were no such a gap, the animacy feature would not be entertained by speakers and would disappear. Therefore, it is likely that the grammar raises this accusative exception as a warning destined to save the animacy feature from oblivion.

6. Po-phrases

In Russian, a numeral phrase takes a distributive meaning when used as complement of the dative preposition *po* (cf. Pesetsky 1982, Franks 1994). A *po*-phrase can be either subject or object of accusative verbs, but surprisingly it cannot be the object of oblique verbs. In addition, the numeral head complement of *po* can be X-specified (i.e. undeclined nominative/accusative) or dative-specified, which is more consistent with the fact that *po* is otherwise a dative preposition. This is illustrated in (62) (Franks 1994) and (63) (Pesetsky 1982).

- (62) a. Každýj učeník polučil po odnomu rublju.
 each student received DIST one.DAT ruble.DAT.SING
- b. Každýj učeník polučil po dva rublja.
 each student received DIST two ruble.GEN.SG
- c. Každýj učeník polučil po pjat' rublej.
 each student received DIST five ruble.GEN.PL
- d. Každýj učeník polučil po pjati rublej.
 each student received DIST five.DAT ruble.GEN.PL
- (63) a. *ja pomogal po devuške v den'.
 I.NOM helped(DAT) DIST girl.DAT.SING in day
 'I helped a girl a day'
- b. *ja upravljaju po zavodu v každom gorode
 I.NOM manage(INST) DIST factory.DAT.SING in each city
 'I manage a (different) factory in each city'

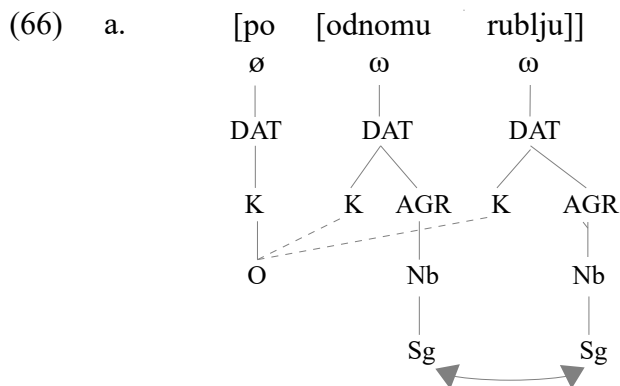
Franks shows that the *po*-phrases in (62) become more puzzling when compared with other dative prepositions. Thus in (64) the numeral phrase complement of preposition *k* is uniformly in the dative, whereas a numeral phrase complement of *po*, other than *odin* (cf. 62a), cannot be in the dative (65a,b). However, a dative numeral is acceptable with a genitive complement, as shown in (65c) (from 62d).

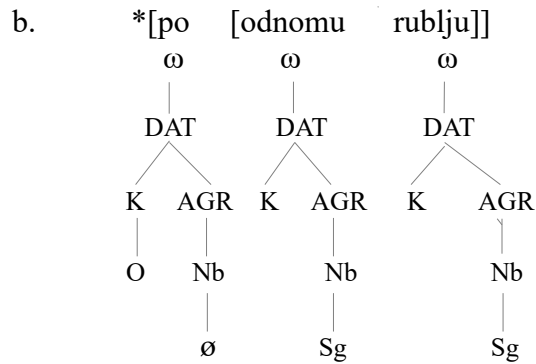
- (64) a. k dvum rubljam
 to two.DAT ruble.DAT.PL
- b. k pjati rubljam
 to five.DAT ruble.DAT.PL

- (65) a. **po dvum rubljam*
DIST two.DAT ruble.DAT.PL
- b. **po pjati rubljam*
DIST five.DAT ruble.DAT.PL
- c. *po pjati rublej.*
DIST five.DAT ruble.GEN.PL

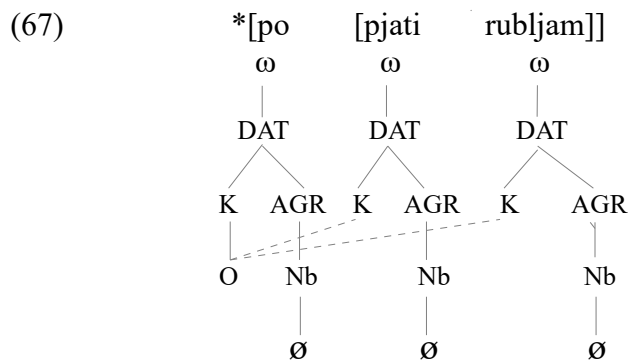
The solution I would like to propose to puzzle out these facts is that *po* represents two distinct but syncretic elements: distributive *po* and prepositional *po*. I assume that distributive *po* governs the dative and is a case assigner (oblique), just like preposition *po*. In addition, distributive *po* has an ω -specified root node, hence is a vector, and it bears a number node with a plural specification [$\emptyset$]. On the other hand, prepositional *po* has a neutral root node and, like other prepositions, it has no AGR node. Thus since distributive *po* has a plural specification, it cannot be used with singular nouns; preposition *po* must be used to fill in the gap.

Under these assumptions, the derivation of the singular *po*-phrase in (62a) proceeds as follows. Singular *po* (the preposition) picks out the constituent of its complement, forcing them to be in the dative, and then the numeral head proceeds to make up the phrase, upholding the dative of its complement, which results in the structure (66a). Notice that *po* spreads oblique case (O) to the K-node of each element in the numeral phrase, and the lines do not cross as the k-nodes and the AGR-nodes belong to different planes. On the other hand, structure (66b) is ill-formed because singular *po* is a better match for the singular numeral phrase. If distributive *po* were used, it would force the numeral phrase to be in the dative plural. Another crucial difference between these structures is that in (66a) *po* is $\emptyset$ -specified in the root node (neutral), while in (66b) it is ω -specified (vector) (see below).





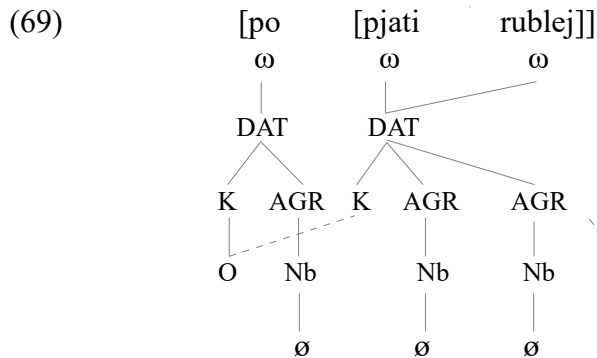
Consider now the higher numeral phrases in (62). Here distributive *po* must be used, as all numerals other than *odin* have a $\emptyset$ -plural specification. Thus *po* selects the constituent that will make the numeral phrase, imposing upon them the dative plural. Then the numeral merges with its complement in the dative plural, which is its best match. However, the resulting structure (67) is ruled out as it contravenes the vector law stated in (68). In effect, the oblique case of vector *po* is assigned to the k-node of the dative noun by overlapping onto the scope of the vector numeral, which is a violation of (68).



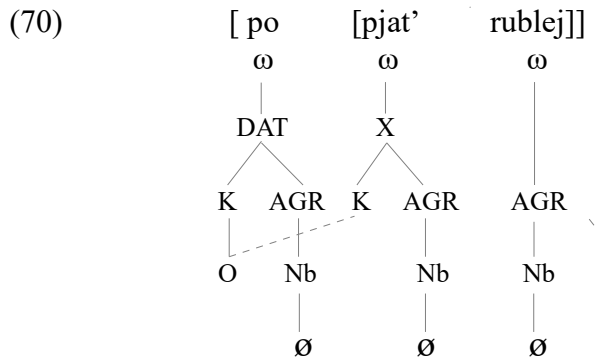
(68) **Vector Barrier Principle**

A vector cannot spread its feature across another vector.

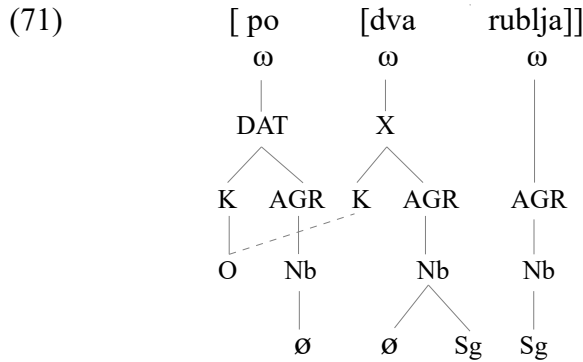
In the present theory, when a structure fails it can be either repaired or replaced by an alternative derivation. In the case at hand, it appears that both options are possible. If the dative numeral forces the noun to be in the genitive, assuming that the derivation iterates, the structure becomes acceptable, as the barrier effect disappears (69).



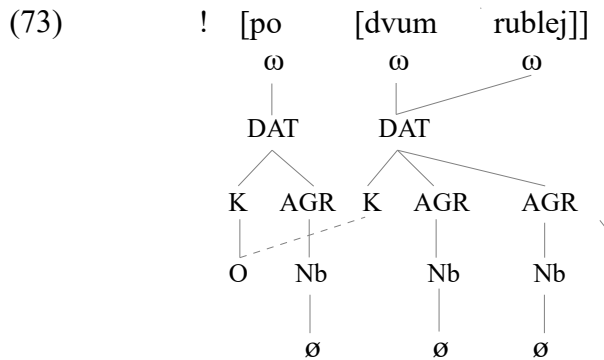
In an alternative derivation, *po* can select the undeclined form of the numeral, which can only merge with a genitive noun, as discussed above. The resulting structure shown in (70) is perfect and even better than (69), which is rather archaic, as Franks points out. The reason for the better acceptability of (70) is that its derivation proceeds straightforwardly from a single input where the numeral has only one option for its complement, unlike (69) where an iteration of the derivation allows to replace the offending dative complement by the genitive, which is obviously a costly operation.



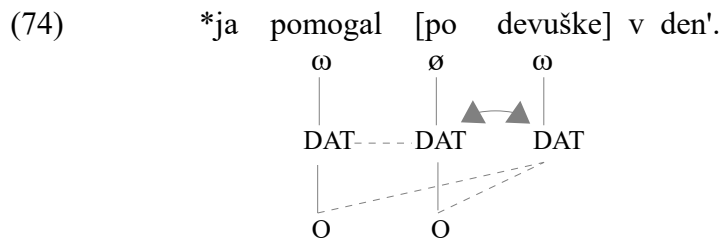
Similarly, lower numerals must be in their X-form with a genitive-marked complement so as to avoid a violation of (68), as shown in (71). In this structure, plural *po* triggers the activation of the recessive plural of the numeral, yielding two number pairs, [$\emptyset - \emptyset$] and [Sg – Sg]. A structure parallel to (69), where the higher numeral is in the dative, is impossible with lower numerals according to Franks. However, in the example he provides (72a), the genitive noun is in singular. It seems that this sentence is excluded anyway because lower numerals govern a singular complement only in their undeclined nominative-accusative form. Pending further information, I assume that (72b), with the noun in genitive plural, should be possible, and it should be archaic as well (the exclamation mark indicates that the acceptability of the sentence is not clear). The representation of (72b) is shown in (73).



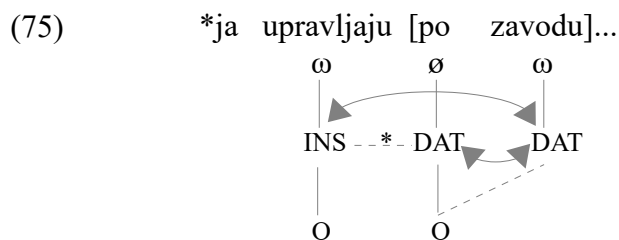
- (72) a. po dva/*dvum rublja
DIST two/two.DAT ruble.GEN.SG
- b. !po dvum rublej
DIST two.DAT ruble.GEN.PL



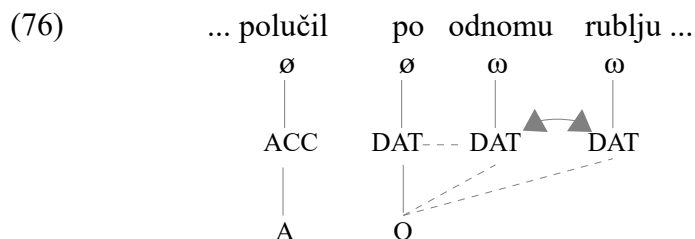
In the examples in (62), the accusative verb cannot interact with the *po*-phrase, and thus its thematic node is stranded. This is because its root node remains in a neutral state ($\emptyset$) instead of mutating into ω , which would make it a vector (see below). On the other hand, oblique verbs are inherently vectors and must interact with the *po*-phrase, causing some unwanted effects. The problem with such verbs, I argue, is due to the ability of *po* to assign oblique case as well as a pendulum effect. The representation of sentence (63a), where preposition *po* is used, is as shown below in (74) (irrelevant features and K-nodes are omitted). In the *po*-phrase, both dative nodes pair up and *po* assigns oblique case to its complement. The dative node of the verb can align with the pair, but its oblique case can neither be absorbed by the oblique case of a case-assigner nor can it spread to *devuške* without inducing a line-crossing effect. Therefore, such a configuration is ill-formed.



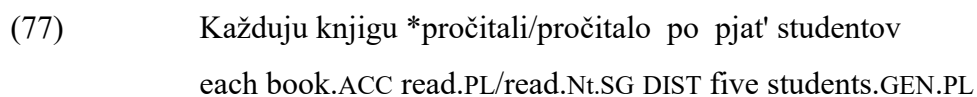
With instrumental verbs, a slightly different scenario arises. As can be seen in (75), *po* and its complement pair up with their dative node. The verb and the complement of *po* group together by a pendulum effect, and thus the instrumental node of the verb must try to align with the dative pair of its mate, yielding a fatal feature clash.



The problem seen in (74) and (75) does not arise with accusative verbs. Such verbs are not inherently vectors, but they are mutable as discussed in Desouvrey (2024). Thus in the following representation of (62a), the neutral accusative verb may not combine with any vector, and hence it cannot align itself with the dative tier. Therefore, its thematic node is stray in another dimension, yielding a well-formed sentence:

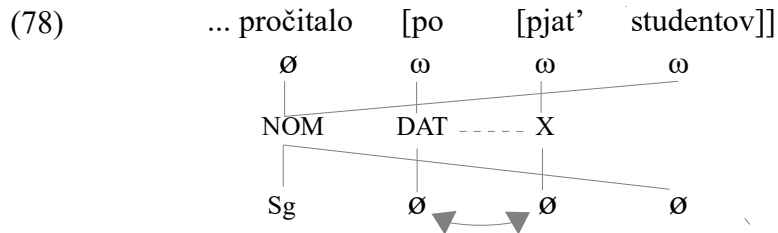


Consider now subject *po*-phrases. According to Franks, the verb must be in the neuter singular, rejecting Pesetsky's claim that *po*-phrases cannot be used as subject, whatever the agreement option. Thus if Franks is correct, (77) is acceptable with the verb taking the neuter singular option. It appears that Franks's view is predicted by our analysis.



‘Five students read each book.’

The representation of (77) is as shown in (78), where irrelevant and non-crucial nodes are conveniently omitted. The dative node of *po* aligns with the X-node of the numeral, allowing their number feature ($\emptyset$) to pair up, while the floating plural of the genitive noun can attach to the verb, which therefore must take its default neuter singular form. Notice that *po* and *pjat'* define a different plan so that there is no line-crossing effect.



7. Conclusion

The key proposal in this paper is that numeral stems are ω -specified in their root nodes, which makes them incompatible with both accusative and nominative morphology. As a result, they are not declined in direct case, and hence they cannot have a thematic node. This unusual state of affairs leads the grammar to raise an exception as a warning to its learners. This paucal warning, which exists only in direct case, is realized by affecting to such numerals a singular feature, forcing the normal underspecified plural feature to recess. Therefore, the complement of such thematic-free numerals can only be in the genitive singular, given that this declension is universally devoid of thematic node. This picture is further complicated by the reappearance of the recessive plural feature when the noun is modified by an adjective, yielding an apparent feature mismatch.

It appears that the agreement process follows from the very nature of the representation. It is shown that identical or compatible features pair up. If all features in the numeral phrase pair up, the verb normally takes its default singular neutral morphology, as it has nothing to agree with. However, if the verb incorporates the features of the genitive noun, it can only agree in plural with a higher numeral head and in singular with a lower numeral head. This analysis of the agreement process makes it possible to link the coreference articulation with the thematic node. It turns out that the R-node is an image of the thematic node, and therefore if an NP has no thematic node, it cannot be the antecedent of an anaphor or a pronoun, irrespective of the agreement form of the verb, which is an epiphenomenon.

This analysis easily carries over to the fact that approximative inversion disallows plural

agreement; it turns out to be simply a matter of line-crossing effect. Moreover, *po*-phrases are accounted for by the same mechanisms once the nature of *po* is identified. It is suggested that there are two instances of *po*: a preposition *po* and a distributive *po*, which are used in singular and plural contexts respectively. The strange fact that *po*, which has a dative thematic node, is incompatible with dative verbs, mainly arises because dative (and other oblique) verbs cannot assign their oblique case to the *po*-phrase without inducing a line-crossing effect in the representation.

Although other languages are not discussed in this paper, I believe that this theory has the power and the flexibility to account for agreement universally. For instance agreement in a related language like Serbo-Croatian is expected to follow from the same principles. Any difference must be imputed to feature inventory. At first glance, it seems that Serbo-Croatian numerals are incompatible with any thematic morphology, which suggest they have an ω -specified stem, while every declension may be either neutral, ϕ -specified, or both.

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